

Visual information is broadcast among cortical areas in discrete channels

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Abstract

Among brain areas, axonal projections carry channels of information that can be mixed to varying degrees. Here, we assess the rules for the network consisting of the primary visual cortex and higher visual areas (V1-HVA) in mice. We use large field-of-view two-photon calcium imaging to measure correlated variability (i.e., noise correlations, NCs) among thousands of neurons, forming over a million unique pairs, distributed across multiple cortical areas simultaneously. The amplitude of NCs is proportional to functional connectivity in the network, and we find that they are robust, reproducible statistical measures, and are remarkably similar across stimuli, thus providing effective constraints to network models. We used these NCs to measure the statistics of functional connectivity among tuning classes of neurons in V1 and HVAs. Using a data-driven clustering approach, we identify approximately 60 distinct tuning classes found in V1 and HVAs. We find that NCs are higher between neurons from the same tuning class, both within and across cortical areas. Thus, in the V1-HVA network, mixing of channels is avoided. Instead, distinct channels of visual information are broadcast within and across cortical areas, at both the micron and millimeter length scales. This principle for the functional organization and correlation structure at the individual neuron level across multiple cortical areas can inform and constrain computational theories of neocortical networks.

eLife assessment

This **important** study uses state-of-the-art, multi-region two-photon calcium imaging to characterize the statistics of functional connectivity between visual cortical neurons. While the evidence supporting the conclusions is **solid**, alternative interpretations of the results cannot be ruled out due to the limitations of calcium imaging, the use of noise correlations as a measure of functional connectivity and putative confounds of behavioural state modulations.

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Introduction

Neurons have characteristic preferences or tuning, which are variables (e.g., stimulus or behavior variables) that correlate with their spiking activity. Neuronal spiking activity is transmitted via axonal projections to other brain areas. In the early stages of visual processing, visual information can be preserved. For example, the retina-to-lateral geniculate nucleus (**LGN**) network tends to preserve unmixed channels, by ensuring that axons from retinal ganglion cells with similar tuning converge on individual LGN neurons (Liang et al., 2018 [DOI](#)). By contrast, the LGN-to-primary visual cortex (**V1**) network famously mixes channels to give neurons receptive fields with both dark-sensing and light-sensing subregions, and thus robust orientation tuning (Hubel and Wiesel, 1962 [DOI](#)). That said, discrete visual information can be transmitted from the retina to the cortex through non-mixing channels. For example, when direction-selective neurons in the retina are genetically ablated, there is a decrease in direction-selective neurons in cortex (Rasmussen et al., 2020 [DOI](#)).

In the visual cortical system in mice, the primary visual cortex (**V1**) and its projections to multiple higher visual areas (**HVAs**) span millimeters (Wang and Burkhalter, 2007 [DOI](#)). Local networks within V1 can have precise local (< 50 microns) cellular-resolution functional connectivity (Ko et al., 2011 [DOI](#)). Studies of longer-range, millimeter-scale networks typically lack cellular resolution, but there are general biases observed. Neurons in V1 and HVAs respond to diverse visual stimuli (Yu et al., 2022 [DOI](#); Vries et al., 2020 [DOI](#)) and are sensitive to a broad range of features including orientation and spatiotemporal frequencies (Andermann et al., 2011 [DOI](#); Marshel et al., 2011 [DOI](#)). Although individual V1 neurons broadcast axonal projections to multiple HVAs (Han et al., 2018 [DOI](#)), the spatiotemporal frequency preferences of these feedforward projections generally match those of the target HVAs (Glickfeld et al., 2013 [DOI](#); Han and Bonin, 2023 [DOI](#); Kim et al., 2018 [DOI](#)). Feedback connections from HVAs carry frequency-tuned visual signals as well (Huh et al., 2018 [DOI](#)). Thus, there are HVA-specific spatiotemporal biases, but cellular resolution and millimeter-scale principles for cortical wiring remain to be elucidated.

In the current study, we investigated the degree of channel mixing between distinctly tuned neurons in the V1-HVA networks of mice by measuring the noise correlations (**NC**, also called spike count correlations Vinci et al. (2016) [DOI](#)) between functional tuning classes of neurons. Functional tuning classes were defined using an unbiased clustering approach (Han et al., 2022 [DOI](#); Yu et al., 2022 [DOI](#); Baden et al., 2016 [DOI](#)). Large field-of-view (**FOV**) calcium imaging enabled us to densely sample across millimeters of cortical space, simultaneously observing large and dense samples of neurons in these tuning classes within and across cortical areas (Yu et al., 2021 [DOI](#); Stirman et al., 2016 [DOI](#)). NCs are due to connectivity (direct or indirect connectivity between the neurons, and/or shared input), and thus provide a trace of connectivity (Cohen and Kohn, 2011 [DOI](#); Vinci et al., 2016 [DOI](#); Snyder et al., 2015 [DOI](#)). In particular, the connectivity that underlies NCs is effective *in vivo*, during normal sensory processing. Thus NCs provide a complement to purely anatomical measures of connectivity. In fact, activity-based estimates of neuronal networks can provide higher fidelity measures than anatomy-based studies (Randi et al., 2023 [DOI](#)). We find that NCs are a reliable measure at the population level. We also find that neuron classes can be categorized into six functional groups, and NCs are higher within these groups (and even higher within classes), both within and across cortical areas, indicating unmixed channels in the network preserve information. Moreover, we find that naturalistic videos draw upon the same functional networks, and modeling suggests that recurrent connectivity rather than bottom-up or top-down input is critical for stabilizing these networks.

Results

Visual cortical neurons form six tuning groups

To measure neuronal activity, we used multi-region population calcium imaging of L2/3 neurons in V1 and four HVAs (lateromedial, **LM**; laterointermediate, **LI**; anterolateral, **AL**; and posteromedial, **PM**) using a multiplexing, large field-of-view two-photon microscope with sub-cellular resolution developed in-house (Stirman et al., 2016) (Figure 1A). Mice expressed the genetically encoded calcium indicator GCaMP6s (Madisen et al., 2015; Chen et al., 2013) in cortical neurons. We located the V1 and HVAs of each mouse using retinotopic maps obtained by intrinsic signal optical imaging (Marshall et al., 2011; Smith et al., 2017) (Figure S1A). We imaged neurons in two to four cortical areas simultaneously (Figure 1A), while mice viewed stimuli on a video display. We typically imaged neurons in V1 and one or more HVAs. Up to 400 neurons (V1: 129 ± 92 ; HVAs: 94 ± 72 ; mean $\pm$ SD; only counting reliably responsive neurons used in subsequent analysis, see Methods) were recorded per imaging region ($500 \times 500 \mu\text{m}^2$). The imaging regions were matched for retinotopy so that the neurons in the simultaneously imaged areas had overlapping receptive fields (RFs). Calcium signals were used to infer probable spike trains for each neuron, as our previous study (Yu et al., 2022). We mapped RFs for individual neurons and populations using small patches of drifting gratings (Figure S1B, C). Neurons in HVAs (LM, AL, PM and LI) had significantly larger RFs than V1 neurons (Figure S1D). Population RFs for a $500 \times 500 \mu\text{m}^2$ imaging region of HVAs covered significantly larger portions of visual space than that of V1 (Figure S1D), as expected given their differing magnification factors (Wang and Burkhalter, 2007; Smith et al., 2017). The overlap of population RFs confirmed that simultaneously imaged cortical areas (V1 and HVAs), each containing ~ 100 neurons, responded to stimuli in the same region of the screen (Figure S1C). These experiments were repeated in 24 mice for a total of 17,990 neurons and NCs were measured for a total of 1,037,701 neuron pairs (Figure T1).

Mouse V1 and HVA neurons exhibit diverse tuning preferences to drifting grating stimuli, in terms of spatiotemporal preferences and sharpness of orientation and direction tuning (Marshall et al., 2011; Andermann et al., 2011; Vries et al., 2020). Previous studies suggested that the axonal projections from V1 to HVAs match the spatiotemporal preferences of the target HVAs (Glickfeld et al., 2013). We sought to determine whether this was a general principle, that extended across V1 and HVAs. We recorded neuronal responses from V1 and multiple HVAs (LM, LI, AL, and PM) to sinewave drifting grating stimuli with various spatiotemporal properties (8 directions $\times$ 3 spatial frequencies $\times$ 3 temporal frequencies for a total of 72 conditions; Figure 1B). HVAs exhibited similar responsiveness and reliability to the 72 different parameterized drifting gratings. V1 and LM were only marginally more reliable than other areas (Figure S1E).

To obtain a granular, data-driven way to classify neurons, neuronal responses were partitioned into 65 tuning classes using an unbiased Gaussian Mixture Model (GMM) (Figure S1F, S2). This GMM classification was reliable, in that the center of the Gaussian profile of each class was consistent among GMMs of random subsets of neurons (Figure 1C). Neurons were consistently classified into the same class (Methods; Figure 1D).

To examine the spatiotemporal frequency selectivity of HVAs, we manually partitioned the 65 GMM classes into six spatial frequency (SF) - temporal frequency (TF) selective groups (Figure 1F). Groups 1, 2, and 3 all prefer low TF (1-2 Hz), and prefer low SF (0.02 cpd), medium SF (0.05 cpd), and high SF (0.19 cpd) respectively. Groups 4, 5, and 6 all prefer high TF (8 Hz) and prefer low SF, medium SF, and high SF respectively. Group 4 (low SF, high TF) was the only group that exhibited increasing responses to the drift speed of the grating stimulus (drift speed = TF/SF, and is measured in deg/s). These groupings were robust and reliable (Figure S1G,H). Similar to the

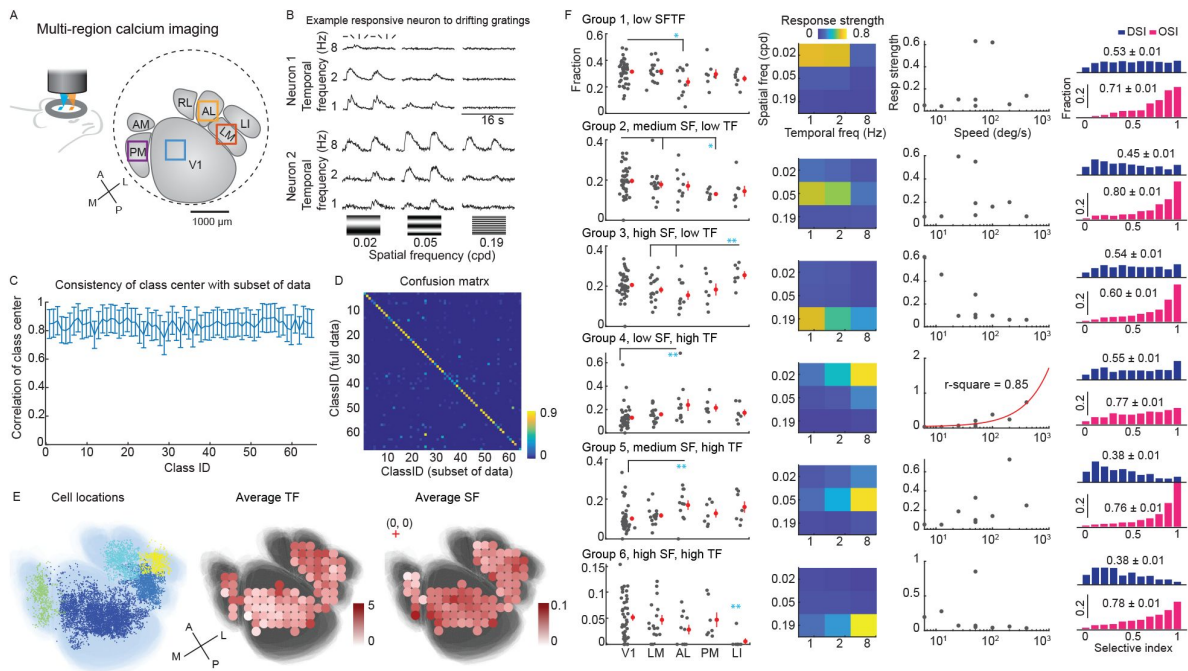


Figure 1.

Functional groups of mouse visual neurons.

(A) Diagram of multi-region two-photon imaging of mouse V1 and HVAs, using a custom wide field-of-view microscope. Example imaging session of the simultaneous recording session of V1, LM, AL, and PM. Squares indicate 500 μm wide imaging regions. (B) Example responses from two neurons (mean calcium trace) to drifting gratings with eight directions at various SF-TF frequencies. (C) Neurons were distributed into 65 different classes using GMM (Fig. S1, S2). The mean correlation coefficients of the center of each class (in principal component space) between GMMs of 10 permutations of a random subset of neurons. (D) The confusion matrix shows that individual neurons are likely (>90%) to remain in the same class even when only a random subset of neurons is used to train the GMM (horizontal), compared to the full data set (vertical). (E) Center of individual neurons (left) overlay on an average visual cortex map. The average visual cortex map was generated by affine registration of visual area maps from all experiments. Neurons are colored by visual areas. Middle, average preferred TF exhibits spatial dependency over the visual cortex (TF: A $\rightarrow$ P, $\text{cor} = -0.25$, $p = 0.015$, M $\rightarrow$ L, $\text{cor} = 0.36$, $p = 0.0004$). Right, the average preferred SF (right) exhibits spatial dependency over the visual cortex (SF: A $\rightarrow$ P, $\text{cor} = 0.35$, $p = 0.0005$, M $\rightarrow$ L, $\text{cor} = -0.06$, $p = 0.54$). Colored dots indicate the average TF and SF (computed with >30 neurons) within patches (180 $\mu\text{m} \times 180 \mu\text{m}$ local areas), overlaid on a map of V1 and HVAs. (F) These 65 classes were manually arranged into six tuning groups based on spatial frequency and temporal frequency (SF-TF) tuning preferences. Column 1, the fraction of neurons in different SF-TF groups. Dots represent individual sessions. Statistical significance was tested by the Ranksum test (*, $p < 0.05$, **, $p < 0.01$). Column 2, the characteristic SF-TF responses of each tuning group. Column 3, speed tuning of tuning groups. Column 4, distribution of cells' orientation selectivity index (OSI) and direction selectivity index (DSI). The number of neurons belonged to the six tuning groups combined: V1, 5373; LM, 1316; AL, 656; PM, 491; LI, 334 (refer to Methods for neuron selection). These six groups provide a compact way of summarizing response diversity, but as shown later, the granularity of the 65 classes provides a superior match to the network properties (Fig. 4F).

previous report, V1 and HVAs have overlapping SF and TF selectivity (Marshall et al., 2011 [↗](#)), with a trend of larger preferred TF from the posterior-medial to the anterior-lateral visual cortex, and a trend of increasing preferred SF from the anterior to the posterior visual cortex (**Figure 1E** [↗](#)). Specifically, AL had a larger fraction of neurons tuned to low SF high TF (Group 4, speed tuning group), and a lower fraction of neurons tuned to low SF-TF (Group 1), compared to V1 (**Figure 1F** [↗](#)). PM had a lower fraction of low TF medium SF neurons compared to V1 and LM (**Figure 1F** [↗](#)). LI had a larger fraction of neurons tuned to high SF and low TF than AL and LM (Groups 3) and had a lower fraction of neurons tuned to high TF and high SF (Group 6) (**Figure 1F** [↗](#)).

Neurons in all six groups exhibited orientation and direction selectivity (**Figure 1F** [↗](#)). The preferred directions of neurons were evenly distributed in V1 and HVAs, except high SF groups (Group 3 and 6) of AL, PM, and LI biased to cardinal directions (**Figure S3A** [↗](#)). The unbiased GMM approach revealed that the orientation selectivity index (OSI) and direction selectivity index (DSI) of visual neurons were jointly modulated by SF and TF (**Figure S3B** [↗](#)). Neurons tuned to high SF and low TF (Group 3) exhibited lower OSI in all tested areas than all of the other groups (Group 3: mean OSI = 0.6; other groups ranged from 0.71 – 0.80; $p < 0.0001$, one-way ANOVA with Bonferroni correction; **Figure 1F** [↗](#), **S3B** [↗](#)). Neurons tuned to high TF and medium-high SF (Groups 5 and 6) exhibited lower direction selectivity than other groups (Group 5, 6, mean DSI 0.38; other groups, mean DSI 0.45 - 0.54; $p < 0.0001$, one-way ANOVA with Bonferroni correction; **Figure 1F** [↗](#), **S3B** [↗](#)).

In summary, we found that neurons in V1 and HVAs are jointly selective to the spatiotemporal frequency and the drifting orientation/direction of gratings. Consistent with (Marshall et al., 2011 [↗](#)), V1 and LI have higher average preferred SF compared to AL ($p < 0.0001$, one-way ANOVA with Bonferroni multi-comparison), while V1 has a lower average preferred TF than all tested HVAs (**Figure S3C** [↗](#)). In contrast to (Marshall et al., 2011 [↗](#)), we found AL has a higher average preferred TF than other visual areas, including LM ($p < 0.0001$, one-way ANOVA with Bonferroni multi-comparison; **Figure S3C** [↗](#)). The discrepancy may be explained by the joint selectivity of spatiotemporal frequency and the orientation/direction, and the different stimuli used in these studies.

NCs are robust measurements of functional networks

A unique aspect of this data set is the scale of the NC measurements, which allows us to measure NCs with individual neuron precision within dense local networks and over millimeter-length scales, in awake mice. Pioneering work in this area focused on local populations, typically less than 1 mm across (Ko et al., 2011 [↗](#); Harris and Mrsic-Flogel, 2013 [↗](#); Lee et al., 2016 [↗](#); Wertz et al., 2015 [↗](#)) or electrode studies over long distances with few neurons in each location (Clay Reid and Alonso, 1995 [↗](#); Siegle et al., 2021 [↗](#)). To investigate the V1-HVA functional network, we computed the NCs of pairs of neurons within individual cortical areas (within-area NC), and NCs for pairs of neurons where the two neurons are in different cortical areas (inter-area NC) (**Figure 2A** [↗](#)). NCs are computed from the residual activity of individual neurons after subtracting the expected neuron firing on nominally identical trials. In this section, we evaluated the fidelity of our NC measurements. We also considered potential bias or noise due to the imprecision of spike inference and the finite number of trials.

We first evaluated the accuracy of NC calculations using inferred spikes from calcium imaging. We characterized the accuracy of spike inference using previously published data of simultaneous two-photon imaging and electrophysiological recording of GCaMP6s-positive neurons from mouse V1 (Chen et al., 2013 [↗](#)). Consistent with a previous benchmark study on spike train inference accuracy (Theis et al., 2016 [↗](#)), we found that the spike train inference methods used in the current study recovered 40-70% of the ground truth spikes (**Figure S4A** [↗](#)). We found that a similar fraction of spikes were missing regardless of the inter-spike interval. Nevertheless, the inferred spike train was highly correlated with the true spike train (**Figure S4A** [↗](#); linear correlation, $r = 0.80 \pm 0.03$ ($n = 6$)). Computing correlations between pairs of neurons using their inferred spike trains accurately reproduced the true correlation values (**Figure S4B** [↗](#); linear correlation, $r = 0.7$).

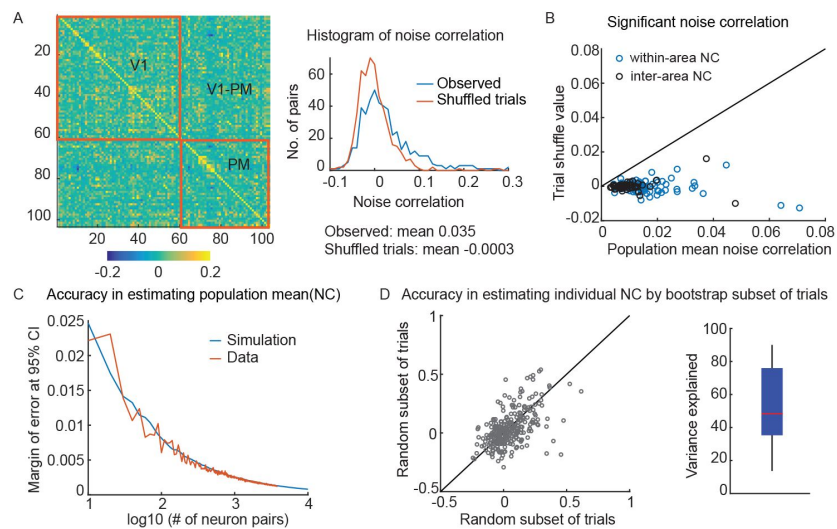


Figure 2.

Noise correlation measurements are reliable

(A) (left) An example covariance matrix of a simultaneous recording of neuronal activity in V1 and PM. (right) NC histogram of an imaging session (blue, note the large positive tail), compared to control, trial-shuffled data (red). (B) The observed population mean NC is always larger than control values (the NC after trial shuffling). Circles indicate the value of individual experiments. (C) Population NCs are precise measures. The margin of error at 95% confidence interval (CI) of the population mean NC reduces rapidly with an increasing number of neuron pairs (72-time bins x 10 trials). With the experimental size of the population (>100 neuron pairs), the estimation precision surpasses the 0.01 level. (D) (left) The NC of individual neuron pairs can be computed using different random subsets of trials, yet reliably converges on similar values (right) The variance of NC computed using a subset of trials is explained by the variance in the held-out subset of trials (53 ± 24 % variance explained; total 204 populations). Each subset contains half of all the trials. The variance explained is defined as the R^2 of the linear model.

We further examined the fidelity of correlation calculations using modified spike trains that are missing spikes. We examined randomly deleting spikes, deleting isolated spikes, or deleting spikes within bursts (**Figure S4D** [↗](#); Methods). We found that at the 1 s time scale, correlation calculations were tolerant to these spike train perturbations. The fidelity of correlation computations was > 0.6 with up to 60% missing spikes (**Figure S4E** [↗](#); Methods). Thus, with conventional spike inference accuracy, about 80% variance of the true correlation is recovered (**Figure S4E** [↗](#)). Thus, NCs are a robust measurement even with imperfectly inferred spike trains.

Next, we evaluated the robustness of NC measurements, given the finite number of trials that are feasible to obtain. We computed NCs for both within-area and inter-area neuron pairs (**Figure 2A** [↗](#)). NCs were computed using spike counts within 1 s bins, similar to previous work with electrophysiology (Cohen and Kohn, 2011 [↗](#); Smith and Sommer, 2013 [↗](#)). Although both within-area and inter-area NCs had wide distributions (range: $-0.2 - 0.6$). The mean NCs across a population were positive and at least five times larger than control data, which are NCs computed after shuffling the trials (5 – 20-fold, 25 – 75% quantile; **Figure 2B** [↗](#)). The estimation of the population mean NC converges fast with increasing numbers of neurons, as suggested by both simulation and experimental data (e.g., the margin of error at 95% CI for mean NC is 0.008 for 100 neuron pairs, **Figure 2C** [↗](#)). While the population-level NC calculations are reliable, the NC estimation of individual neuron pairs is noisier due to the limited number of trials, albeit positively correlated (**Figure 2D** [↗](#)). A linear model explains about $53 \pm 24\%$ of the variance between NCs for individual neuron pairs computed using different random subsets of trials. In summary, this evidence indicates that NCs can be accurately measured at the population level with our large FOV calcium imaging methods, despite imperfect spike train inference and a finite number of trials.

Tuning similarity is a major factor in the V1-HVA functional network

Having established that NC measurements are reliable and robust at the population level, we examined potential NC-regulating factors, including firing rate (joint across the pair), the physical distance between the neurons (laterally, across the cortex), signal correlation (SC, the similarity between two neurons' average responses to stimuli), and RF overlap. We assessed the contributions of individual factors using a linear model. We found that both within- and inter-area NCs are similarly modulated by the aforementioned factors (**Figure 3A** [↗](#); r^2 of the linear regression model). SC is the most pronounced factor that explains about 10% of the variance of within-area NCs, and about 5% of the variance of inter-area NCs (**Figure S5A** [↗](#)). The fraction of RF overlap contributes about 6% and 3% variance of within- and inter-area NCs, respectively (**Figure S5B** [↗](#)). The firing rate explained about 2% of the variance of within- and inter-area NCs (**Figure S5C** [↗](#)). The cortical distance explained about 2 – 3% of the variance of the NCs (**Figure S6** [↗](#)).

SC, RF overlap, and firing rate positively regulate both within- and inter-area NC, with SC providing the strongest predictor. Cortical distance negatively regulates the within-area NC (**Figure S6A** [↗](#)), as expected. However, cortical distance is non-monotonically related to the inter-area NC (**Figure S6B** [↗](#)). This can be explained by how retinotopic organization progresses across area boundaries. That is, when the RF locations are accounted for, a monotonic decrease in NC with distance can be recovered (**Figure S6C-H** [↗](#)).

We then evaluated whether NCs are modulated by tuning similarity independent of RF overlap. In the subset of orientation-selective neurons, both within- and inter-area NCs were significantly modulated by orientation-tuning selectivity. That is, neuron pairs that shared the same preferred orientation exhibited higher NCs (**Figure S5D** [↗](#)). NCs of a subset of neurons with non-overlapping RFs were significantly higher when the neurons shared the same preferred orientation (**Figure**

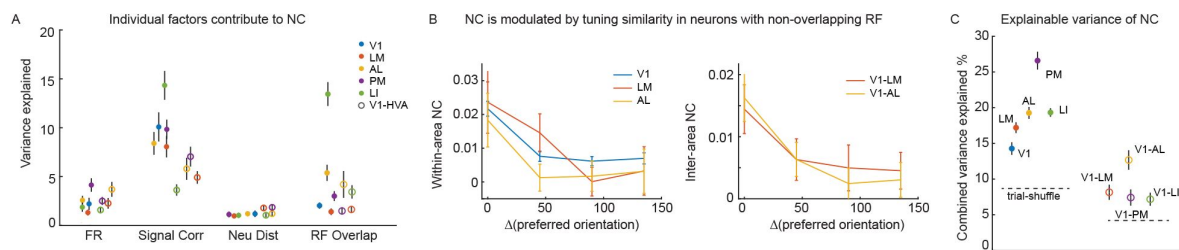


Figure 3.

Factors that contribute to mesoscale NC.

(A) The variance of within-area and inter-area NCs (during grating stimuli) is explained by individual factors including firing rate (FR), signal correlation (signal corr.), neuron distance (neu. dist.), and receptive field (RF) overlap. (B) NCs of neurons with non-overlapping RF is modulated by orientation tuning similarity (within-area, $P_{V1} < 10^{-4}$ (N = 3401), $P_{LM} = 0.03$ (N = 181), $P_{AL} = 0.019$ (N = 284); inter-area: $P_{V1-LM} = 0.019$ (N = 650), $P_{V1-AL} = 0.0004$ (N = 998); t-test). (C) The variances of within-area NCs and inter-area NCs are explained by FR, signal corr., neu. dist., and RF overlap combined. The variance explained is a multi-linear regression model's R^2 . (A, C) The error bar indicates the standard error of the mean of permutations. A subset of 100 neuron pairs was randomly selected for each permutation.

3B [↗](#); t-test, $p < 0.05$ for V1, LM and AL neuron pairs, insufficient data for PM and LI). This result confirms that the connectivity between neurons is modulated by tuning similarity (SC) independent of RF overlap, over millimeter distance scales.

Overall, about 20% of the variance of within-area NCs, and 10% of the variance of inter-area NCs are explained by the aforementioned factors jointly (**Figure 3C** [↗](#); r^2 of multi-linear regression model). Although inter-area NCs have a smaller mean and variance, it is less predictable by known factors (within-area NCs pooled over all tested area 0.012 ± 0.052 , inter-area NCs between V1 and all tested HVAs 0.0063 ± 0.04 ; both t-test and F-test $p < 10^{-4}$). In an expansion of prior work on local functional sub-networks (Lee et al., 2016 [↗](#); Wertz et al., 2015 [↗](#); Ko et al., 2011 [↗](#); Harris and Mrsic-Flogel, 2013 [↗](#)), we find that signal correlation is the strongest factor regulating both within-area and inter-area NC networks, suggesting that neurons exhibiting similar tuning properties are more likely to form functional sub-networks across a broad spatial scale, spanning millimeters in the mouse V1-HVA network.

Neurons are connected through functionally distinct, unmixed channels

Since HVAs exhibit biased SF-TF selectivity (**Figure 1E** [↗](#)), and tuning selectivity (i.e., SC) is a major factor for functional connectivity even across the millimeter length scale (**Figure 3A** [↗](#)), we assessed the precision of this network in the tuning (SC) dimension. We performed additional analysis to determine whether the ST-TF biases in HVAs could be due to simple, weak biases in the NC connectivity. Alternatively, there could be non-mixing channels of connections in the V1-HVA network to preserve information among similarly tuned neurons. We found evidence for this latter situation. Moreover, we found that the non-mixing channels consist of a greater number of neuron pairs with high NCs.

For this analysis, we focused on neuron pairs with high NCs, which we defined as NCs > 2.5 standard deviations above the control (trial-shuffled) NCs for the population (**Figure 4A** [↗](#)). We focused on these high NC pairs because they can represent high-fidelity communication channels between neurons. Within each SF-TF group, for both V1 and HVAs, about 10-20% of neuron pairs exhibited high NCs, in contrast to 5% for inter SF-TF group connections (**Figure 4A, B** [↗](#)). The fraction of pairs that exhibit high NCs is relatively uniform across tuning groups and HVAs with a few exceptions (**Figure 4B** [↗](#)). For example, in HVA PM group 3 contains a higher fraction of high fidelity connections than the other HVAs. Overall, these results show that mixing between groups is limited, and instead, group-specific high-NC sub-networks exist between neurons across millimeters of cortical space.

Prior findings from studies of axonal projections from V1 to HVAs indicated that the number of SF-TF-specific boutons underlies the biased frequency tuning in LM, AL, and PM, while the strength of a small fraction of speed tuning boutons contributes to the biases in speed tuning among these HVAs (Glickfeld et al., 2013 [↗](#)). Though the functional connectivity is not completely defined by the feedforward axonal projections from V1 (Huh et al., 2018 [↗](#)), this number vs. strength question is one that we can address with our data set. We found that the biased representation of SF-TF among HVAs is linearly related to the number of neuron pairs with high NCs (**Figure 4C** [↗](#)). By contrast, the average NC value (mean strength of connection) for each tuning group is not linearly related to the fraction of SF-TF groups in each HVA (**Figure 4D** [↗](#)). That is, the biasedly represented tuning group in HVA does not tend to have a higher functional connectivity with V1. Thus, the biases in SF-TF representation are likely related to the abundance of significant SF-TF-specific connections, but not the strength of the connections.

To this point, we have focused on the six SF-TF groups. The evidence supports group-specific channels among these neurons. However, these six groups originated with 65 *classes* from data-driven GMM clustering and were manually collected into the six SF-TF *groups* (**Figure 1** [↗](#)). The

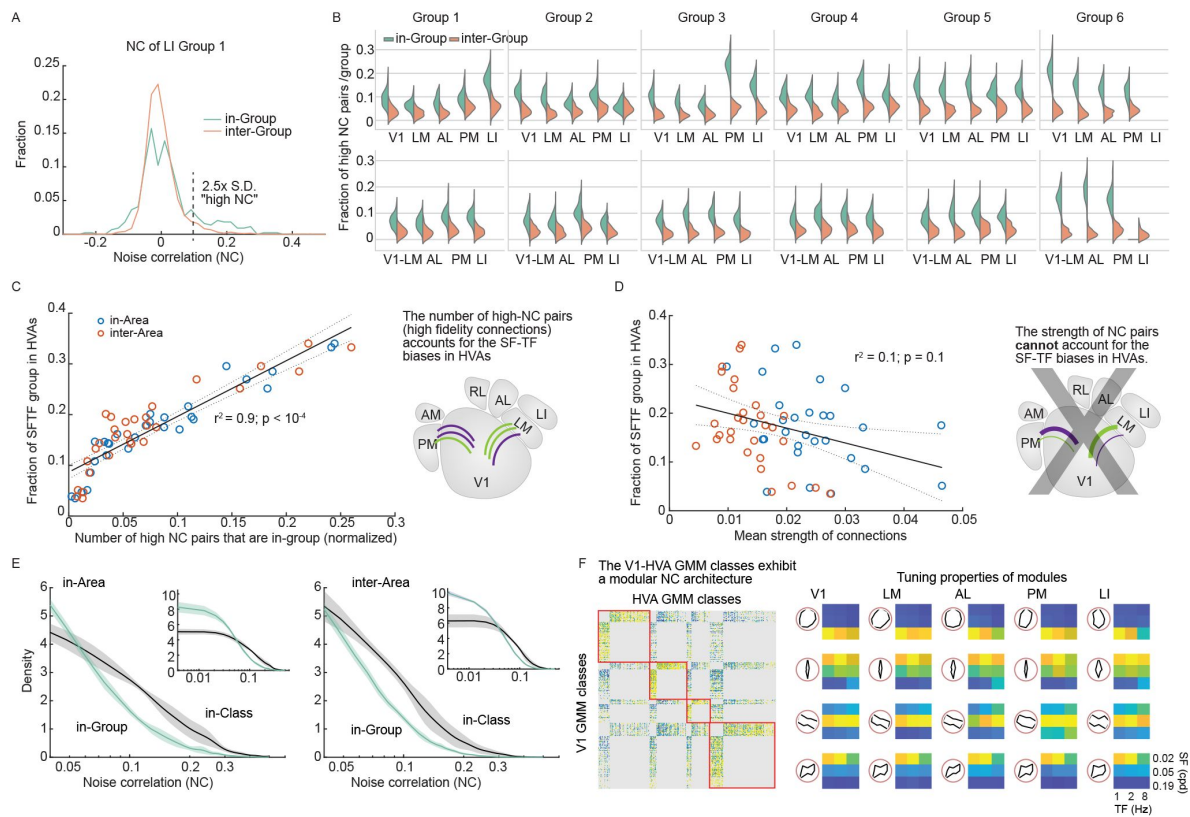


Figure 4.

High-fidelity tuning-specific V1-HVA communication channels

(A) The distribution of NCs of a subset of LI neurons from tuning group 1 (blue) and NCs of a subset of LI neurons between group 1 and other tuning groups (red). In the next panel, we will focus on the positive tail: the portion of the distribution that is over 2.5x of the S.D. **(B)** Fraction of neuron pairs with high NCs ($> 2.5 \times$ S.D. of trial-shuffled NCs) for within-group and inter-group pairs. Neurons within a group have a larger fraction of neuron pairs exhibiting high-fidelity connections (all comparisons, t-test, $p < 0.0001$). Distributions were generated with 100 permutations. **(C)** The fraction of high-fidelity connections is linearly related to the fraction of SF-TF groups in each HVA ($r^2 = 0.9$; $p < 0.0001$). The X-axis indicates the number of high NC neuron pairs in the simultaneously imaged V1-HVA populations divided by the total number of high NC pairs in the V1-HVA population imaged (sum over all groups). *(right)* This result is summarized in a diagram indicating that area-specific SF-TF biases correlate with the number of high-fidelity functional connections. **(D)** The average NC value (mean strength of connection) for each tuning group is not linearly related to the fraction of SF-TF groups in each HVA ($r^2 = 0.1$; $p = 0.1$). *(right)* This result negates the hypothesis suggested by the diagram, where area-specific SF-TF biases correlate with the strength of functional connections. Instead, the number of connections (panel C) seems to account for the observed trends. **(E)** Density function plots of NCs for in-area (left) or inter-area (right) neuron pairs that shared the same GMM-based class (65 classes) or group (six SF-TF preference groups) indicate that the more granular, GMM-based class categorization accounts for the structure of the NC network with higher fidelity than the coarser SF-TF groups (full scale is inset, bin size is 0.00875). **(F)** The functional connectivity matrix for the V1-HVA network between GMM classes exhibits a modular structure. *(right)* Each module has a particular tuning selectivity and SF-TF bias. The value of this functional connectivity matrix was the fraction of high NC pairs.

trends we see for groups may reflect general SF-TF biases. In that case, we would expect that the in-class NCs would exhibit similar distributions of NCs as the in-group NCs. However, there might be further precision in the specific channels not captured by the SF-TF groups. A hint towards that can be seen in the fact that orientation tuning can modulate NCs (**Figure 3**). Indeed, when we plotted the NC distribution for in-class neuron pairs and compared it to the distribution for in-group neuron pairs, we found a pronounced positive tail for the in-class distribution (**Figure 4E**). Thus the GMM classes provide relevant, granular labels for neurons, which form functional sub-networks with non-mixing channels, which are more precise than predicted from coarse SF-TF biases or groups.

The GMM classes are widely distributed in all tested areas (**Figure S7B**). We constructed an inter V1-HVA connectivity matrix for the 65 classes (**Figure 4F**). The connection weight is determined by the proportion of pairs exhibiting high NCs. To investigate the modular structure of this network, we performed community detection analysis using the Louvain algorithm (Rubinov and Sporns, 2010). This analysis assigned densely connected nodes to the same module (**Figure 4F**). Overall, the connectivity matrix was split into four community modules (**Figure 4F**; **S7C**). Interestingly, the corresponding nodes in V1 and HVAs within each community module have similar direction and SF-TF preferences (**Figure 4F**). For example, the module 2 nodes exhibited narrow vertical direction tuning and preferred high SF and low TF. Module 1 exhibited high SF preference without direction bias. Area differences in the characteristic tuning selectivity of each module are small, suggesting that the GMM class channels are common across the V1-HVA network. This is consistent with the overall broadcasting projection structure of V1 neurons (Han et al., 2018).

In summary, V1 and HVA neurons can be classified by their selectivity to oriented gratings, and they form precise, discrete channels or sub-networks. These sub-networks of neuron pairs with high NCs preserve selectivity by limiting inter-channel mixing. The organization of V1-HVA sub-networks exhibited properties consistent with those of V1-HVA feedforward projections, where the number of high-fidelity connections, rather than the strength of the connections, accounted for SF-TF biases among HVAs. Moreover, the precision of these networks extends beyond prior observations of general SF-TF biases, to include orientation and direction tuning.

Functional connectivity is stable across stimuli

Functional connectivity can be dynamic and transient, which complicates its relationship with structural (i.e., anatomical) connectivity, yet can provide more accurate predictions for network dynamics than the latter (Randi et al., 2023). We performed additional analysis to determine whether the NC-based functional connectivity analysis we performed above provides fundamental in-sights into neuron circuits beyond a stimulus-specific transient. We compared NC measurements in response to drifting gratings (NC_{grat}) to NC measurements in response to naturalistic videos (NC_{nat}). This analysis was restricted to the subset of neurons that responded to both types of stimuli in a separate set of experiments. So far, we have shown that SC (i.e., neuron tuning similarity) is the best predictor for NCs. However, a neuron pair that shares a high SC to drifting gratings does not guarantee a high SC to naturalistic videos ($corr(SC_{grat}, SC_{nat}) = 0.084 \pm 0.065$). Thus, it is reasonable to expect that NCs in response to gratings do not predict the NCs in response to naturalistic videos. However, we were surprised to find that the correlation between NCs to the two stimuli is significantly higher than that of the corresponding SCs ($corr(NC_{grat}, NC_{nat}) = 0.22 \pm 0.13$; **Figure 5A**). Thus, NCs across stimulus types are more predictable than SCs across stimulus types. To our knowledge, this is the first time this has been reported.

We used SCs to naturalistic videos (SC_{nat}), and gratings NC_{grat} to predict NC_{nat} using linear regressions. Both predictors are positively related to the NC_{nat} (**Figure 5B**). We found that NC to gratings outperformed SC to naturalistic videos in predicting NC to naturalistic videos (t-test, $p < 0.0001$; **Figure 5C**). Meanwhile, combining both predictors adds almost linearly in predicting NC to natural videos (**Figure 5C**), suggesting that the cross-stimulus NC predictor adds an

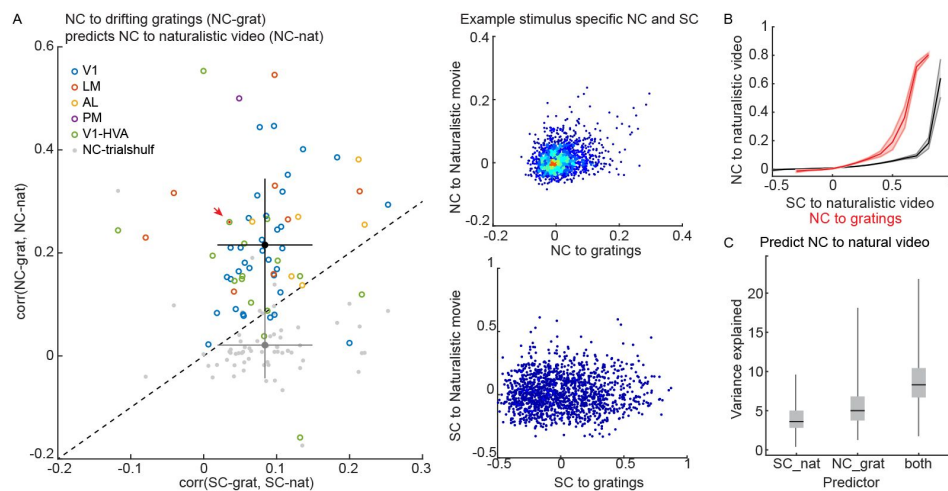


Figure 5.

Noise correlations (NCs) across different classes of stimuli are more stable than tuning, or signal correlations (SCs).

(A) (left) NCs measured during the naturalistic video are well correlated with NCs measured during drifting grating stimuli. The correlation between NCs across different stimuli is significantly higher than the correlation between corresponding SCs ($\text{corr}(NC_{\text{grat}}, NC_{\text{nat}}) = 0.22 \pm 0.13$; $\text{corr}(SC_{\text{grat}}, SC_{\text{nat}}) = 0.084 \pm 0.065$; t-test, $p < 0.0001$). Colored circles represent individual experiments. Gray dots represent trial-shuffled control ($\text{corr}(NC_{\text{shuf}}_{\text{grat}}, NC_{\text{shuf}}_{\text{nat}}) = 0.02 \pm 0.06$). The black/gray dot and error bars indicate the mean and SD for NC and SC. (right) The correlation between (top) NCs and (bottom) SCs during grating and naturalistic video stimuli in an example dataset (red arrowhead in the lefthand plot). (B) NCs to a naturalistic video are positively related to the SCs, as well as to the NCs to drifting gratings. The shaded area indicates SEM. (C) The percentage of NC_{nat} variance is explained by a linear model of SC_{nat} , NC_{grat} , or both factors. NC_{grat} is a better linear predictor compared to SC_{nat} (NC_{grat} $5.3 \pm 3\%$; SC_{nat} $4 \pm 2\%$; t-test, $p < 0.0001$). Combining both factors predicts the NC_{nat} even better ($8 \pm 3\%$; t-test, $p < 0.0001$). Variance explained is measured by R^2 of the linear regression.

independent dimension to the SC predictor. These results are evidence that NC-assessed functional connectivity reflects a fundamental aspect of the architecture of neuronal circuitry that is independent of visual input.

Functional connectivity is not explained by the arousal state

Previous studies in awake mice suggested that a significant portion of the single-trial variance of visual cortical neural activity can be attributed to implicit behavioral or arousal factors during spontaneous locomotion activity (Stringer et al., 2019 [↗](#); Dadarlat and Stryker, 2017 [↗](#)). We wondered whether the cross-stimulus stability of NCs observed in the current study would be explained by top-down modulation or the arousal state of the animal. The arousal state of the animal can be assessed using pupil dynamics (Reimer et al., 2014 [↗](#)). Here we characterized the arousal state by a multidimensional matrix composed of pupil centroid, pupil area, and principal components of pupil video (Methods; **Figure S8A** [↗](#)). We measured the contribution of the multidimensional arousal factors to the variance of population neural activity.

We found that neuron activity is highly stimulus-driven while the animal is in a quiet state of wakefulness. The stimulus accounted for $30 \pm 11\%$ of the single trial variance of neural activity. Meanwhile, the multidimensional arousal factors explained about $3.5 \pm 2\%$, which is a significantly smaller contributor than the stimulus ($p < 0.0001$, paired t-test; **Figure S8B** [↗](#)). The visual stimulus is more than ten times more influential over the trial-to-trial variability in activity than the arousal factors (**Figure S8C** [↗](#)).

Moreover, independent of the arousal factors and the visual stimulus, about $6 \pm 4\%$ of the activity of each neuron can be predicted by neuronal activity in its local population (greater than the contribution from arousal factors; $p < 0.0001$, paired t-test; Methods; **Figure S8B** [↗](#)). We reproduced the cross-stimulus comparison of SCs and NCs (**Figure 5A** [↗](#)) using just the residual neural activity after subtracting the arousal-modulated portion. The cross-stimulus stability of NC connectivity is preserved in the residual neural activity (**Figure S8D** [↗](#)). These results provided additional evidence that the cross-stimulus stable NC network relies on the circuitry of the neuron population rather than the bottomup stimulus-driven activity or top-down modulation.

Notice that the contribution of arousal factors to population neural activity in the current dataset is smaller than the reported value from mice engaged in locomotion activity (Stringer et al., 2019 [↗](#)). In one study (Stringer et al., 2019 [↗](#)), a multidimensional arousal and facial dynamics matrix accounted for 21% of the variance of spontaneous neural activity. The difference could be due to the different brain states of the mice. In the current study, mice were in quiet wakefulness. The fluctuation of the arousal state during quiet wakefulness is much smaller than that during locomotion activity (Reimer et al., 2014 [↗](#)).

Recurrent connection contributes to the stability of NC network

After we observed the surprising cross-stimulus stability of the NC-based functional connectivity, we investigated potential underlying mechanisms. NCs can be due to both shared input (Shadlen and Newsome, 1998 [↗](#)) and direct/indirect wiring (Doiron et al., 2016 [↗](#)). Indeed, using a simple model with two leaky integrate-and-fire (LIF) neurons, we found that the NC is positively regulated by a larger fraction of shared input as well as by the increasing recurrent connection strength (**Figure 6A, B** [↗](#)). We then asked how the two sources contribute to the cross-stimulus stability of the NC functional network using LIF neuronal network simulations (**Figure 6C** [↗](#)). The simulated neuronal network contains 80 excitatory neurons and 20 inhibitory neurons that are randomly connected. The input layer contains 1000 independent Poisson spiking neurons. The network parameters are determined based on previous work (Song et al., 2000 [↗](#)) and all the simulations generated comparable LIF firing rates (4-6 Hz), as well as NCs (population mean: 0.05-0.25) and SC values (population mean: 0.01-0.15).

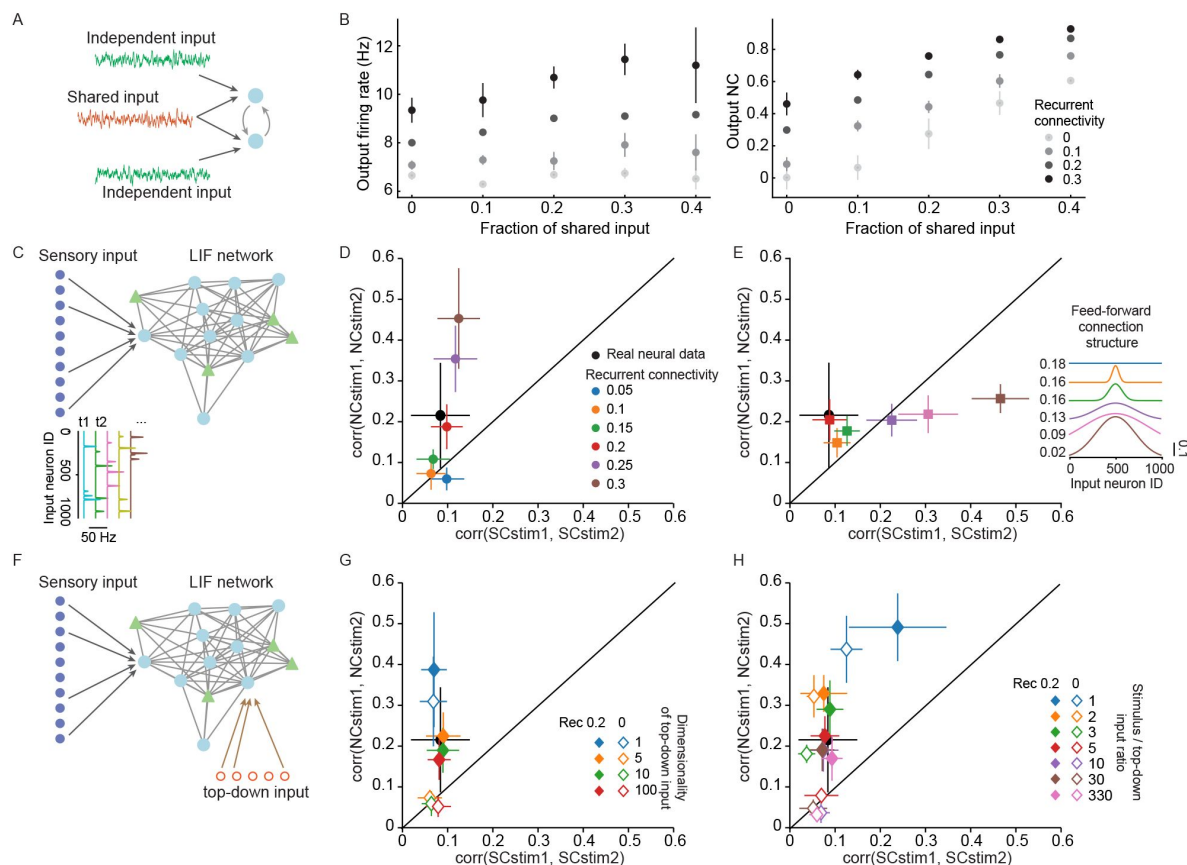


Figure 6.

A network simulation shows that recurrent connectivity can contribute to the stability of the NC network.

(A) A model with two leaky integrate and fire (LIF) neurons that are connected through excitatory synapses. The LIF neurons receive a fraction of shared input (red) and independent input (green) from a Poisson input layer. (B) The firing rate (left) and NC (right) of the two LIF neurons in a toy model (A) is regulated by the fraction of shared input and the strength of the recurrent connection. (C) Schematic of an LIF neuron network model with randomly connected LIF neurons and an input Poisson layer. The structure of the input connection and the strength of the recurrent connection are modulated in the simulation (D, E). (D) In networks with random input connection structures, increasing recurrent connection strength leads to higher cross-stimulus stability of the NC network. Among the values tested, recurrent connection = 0.2 (red) generated a network that was closest to the mouse L2/3 visual neurons (black). (E) In simulations with 0.2 recurrent connectivity strength, regulating the input structure does not change the cross-stimulus stability of the NC network but leads to higher cross-stimulus stability of the SC network. (F) Same LIF network as C with an additional source of top-down input. The dimensionality and strength of the top-down input are modulated in the simulation (G, H). (G) In a network that is mainly driven by sensory input and receives moderate top-down modulation, the sensory/top-down input ratio is 10, recurrent connection is required to reproduce the NC network's cross-stimulus pattern. Changing the dimensionality of top-down input generated similar NC network patterns (overlapping dots in the figure). (H) In networks that receive strong top-down input, the sensory/top-down input ratio is no greater than 1.6, top-down modulation alone can generate cross-stimulus stable NC. Error bars in all panels indicate the SD of multiple randomly initialized simulations under the same condition.

In the first set of simulations, the feedforward (FFD) connection from the input layer to the LIF network is random. Increasing recurrent connection strengths (ranging from 0.05 to 0.3) generated NC-based networks with higher cross-stimulus stability (**Figure 6D** [↗](#)). A recurrent connection strength of 0.2 best reproduced the mouse data. In the second batch of simulations, we fixed the recurrent connection strength to 0.2 but manipulated the input FFD connection structure ranging from random to increasingly wider bell shapes (**Figure 6E** [↗](#)). This means that the local neurons received increasingly similar FFD input. We found that increasingly similar local FFD input did not lead to higher NC stability, but did increase SC similarity across stimuli (**Figure 6E** [↗](#)). Also, the random FFD input connection structure (0.18 FFD, red) reproduced the experimentally observed NC network the best (**Figure 6E** [↗](#)). Thus, these LIF simulations showed that although both shared input and recurrent connections contributed to the NC, the recurrent connections are critical for generating the observed cross-stimulus stability of the NC functional network.

As top-down modulation was suggested to contribute to neural activity, we introduced additional inputs to the LIF network to explore the interplay between top-down inputs and NC-based network structure (**Figure 6F** [↗](#)). We compared the cross-stimulus stability of NC and SC in top-down modulated LIF networks with or without recurrent connections. When the LIF network receives moderate strength top-down inputs and strong stimulus input, recurrent connectivity is required to reproduce the cross-stimulus stable NC connectivity for multi-dimensional top-down modulation. Note that in the present data set, we found that the stimulus/top-down input ratio is about 10 (**Figure S8C** [↗](#)). This is true for cases when the top-down input is multi-dimensional (**Figure 6G** [↗](#)), and both the current study and a previous study found (Stringer et al., 2019 [↗](#)) that indeed top-down input to the mouse visual cortex is high dimensional (**Figure S8B** [↗](#)). On the other hand, when top-down input is unrealistically strong (the stimulus/top-down input ratio is no greater than three, instead of the ten-fold factor we find in our data), the LIF network can generate crossstimulus stable NC connectivity with or without recurrent connections (**Figure 6H** [↗](#)). However, as stated above, in the current study, the visual cortical neuron population is highly stimulus-driven (10-fold greater than top-down input; **Figure S8C** [↗](#)). Together, these LIF simulations suggested that recurrent connections are critical for generating the observed cross-stimulus stable NC connectivity.

Discussion

We used large-scale two-photon calcium imaging across cortical areas to show that neuron-resolution, NC-based assessments of functional connectivity exhibited tuning-specific organization, across millimeter length scales. We provided a detailed analysis of the rigor of measuring NCs with calcium imaging, and assessed their reliability given the imprecision of calcium imaging for inferring spiking activity. The connectivity we observed reinforces data from axonal projection patterns (Glickfeld et al., 2013 [↗](#); Han et al., 2018 [↗](#); Kim et al., 2018 [↗](#)), and provides more granular functional resolution, down to >60 tuning classes we found with GMM analysis. Thus, V1 broadcasts high-fidelity channels of information to HVAs. The projections preserve fidelity by minimizing the mixing among tuning channels. Such a connectivity pattern readily supports the generation of segregated tuning biases in mouse HVAs (Yu et al., 2022 [↗](#); Marshel et al., 2011 [↗](#)). However, it remains unclear whether this supports the generation of new tuning features in HVAs that do not exist in V1, which would require feature integration (Juavinett and Callaway, 2015 [↗](#)). Thus, further studies in anatomical and functional connectivity are needed to address the issue of feature integration.

NCs are an activity-based trace of connectivity in neuronal networks. NCs can provide specific insights about neuronal circuit architecture when analyzed rigorously (Cohen and Kohn, 2011 [↗](#); Schulz et al., 2015 [↗](#); Ecker et al., 2014 [↗](#); Doiron et al., 2016 [↗](#)). For example, researchers have proposed quantitative hypotheses for establishing a link between NCs and information encoding (Kohn et al., 2016 [↗](#); Moreno-Bote et al., 2014 [↗](#); Kanitscheider et al., 2015 [↗](#); Romyantsev et al.,

2020 [\[link\]](#); Hazon et al., 2022 [\[link\]](#); Kafashan et al., 2021 [\[link\]](#)), and information transmission (Zandvakili and Kohn, 2015 [\[link\]](#); Ohiorhenuan et al., 2010 [\[link\]](#)), as well as neuron tuning (Ecker et al., 2011 [\[link\]](#); Panzeri et al., 2022 [\[link\]](#)). NCs offer a data-driven complement to model-driven driven approaches for analyzing neural circuitry (Pillow et al., 2008 [\[link\]](#); Keeley et al., 2020 [\[link\]](#); Goris et al., 2014 [\[link\]](#); Rabinowitz et al., 2015 [\[link\]](#)), and approaches that require cellular resolution manipulations (Randi et al., 2023 [\[link\]](#); Oldenburg et al., 2024 [\[link\]](#)). Technological advancements continue to enable further access to anatomical network structures (Turner et al., 2022 [\[link\]](#); Gao et al., 2022 [\[link\]](#); Velicky et al., 2023 [\[link\]](#)), but since it is difficult to predict the emergent behavior of a collection of neurons from purely anatomical information, functional connectivity, or NC in particular, provides a valuable bridge.

To aid in generating a mechanistic interpretation of NC, one strategy is to compare NC across different states (Doiron et al., 2016 [\[link\]](#)), such as anesthetized versus awake (Ecker et al., 2014 [\[link\]](#)), walking versus stationary (Dadarlat and Stryker, 2017 [\[link\]](#)), spontaneous versus stimulus-driven conditions (Miller et al., 2014 [\[link\]](#)), and different task conditions (Ruff and Cohen, 2016 [\[link\]](#)). In the current study, we compared the NC of the mouse visual cortex to two different visual stimuli. Our findings of cross-stimulus stability of NC provide the first evidence showing that NC outperforms SC in predicting the functional neural network to a different visual stimulus in the mouse visual cortex. It is encouraging that despite limitations in NC measurement and uncertainty regarding its source, the NC-SC relationship is stable and provides effective constraints to neuron network models with stimulus input, recurrent connections, and multi-dimensional top-down modulation. NC characterizes the activation structure of a functional neuron network. Ensemble analysis serves a similar purpose by directly identifying a subset of coactivated neurons. Previous ensemble analyses have suggested that the co-firing patterns of neurons in response to a visual stimulus are highly similar to those observed during spontaneous co-firing events (Pérez-Ortega et al., 2021 [\[link\]](#); Miller et al., 2014 [\[link\]](#)). Both the ensemble analysis and the NC analysis emphasized the critical role of intrinsic network interactions, rather than bottom-up or top-down inputs, in generating the emergent behavior of neuron populations.

Recent breakthroughs in systems neuroscience have been made possible by advancements in large-scale population neuron recording techniques (Yu et al., 2021 [\[link\]](#); Stirman et al., 2016 [\[link\]](#); Papadopoulou et al., 2024 [\[link\]](#); Manley et al., 2024 [\[link\]](#); Siegle et al., 2021 [\[link\]](#)). Interpreting the circuit mechanisms underlying the collective behavior of large neuron populations is challenging but offers significant opportunities for understanding the brain (Urai et al., 2022 [\[link\]](#)). In this study, we demonstrated that characterizing the functional connectivity, specifically the noise correlation (NC), of the neuron population provides valuable insights into interpreting the underlying circuit mechanisms. However, a quantitative model linking functional connectivity with the emergent activity of the neuron population is still missing. This motivates future studies in computational modeling and experimentally probing the emergent activity of neuron populations using behavior or artificial methods.

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Author Contributions

All experiments and analyses were performed by YY. The imaging system was built by JNS. Analysis was assisted by CRD. Study design and supervision by SLS. YY and SLS wrote the paper.

Disclosures

SLS serves as a consultant for optics and neuroscience companies and is a founder of Pacific Optica.

Methods

Animals and surgery

All animal procedures and experiments were approved by the Institutional Animal Care and Use Committee of the University of North Carolina at Chapel Hill or the University of California Santa Barbara and carried out in accordance with the regulations of the US Department of Health and Human Services. GCaMP6s expressing transgenic adult mice of both sexes were used in this study. Mice were 110 - 300 days old for data collection. GCaMP6s expressing were induced by the triple crossing of TITL-GCaMP6s line (Allen Institute Ai94), Emx1-Cre line (Jackson Labs #005628), and ROSA:LNL:tTA line (Jackson Labs #011008) (Madisen et al., 2015 [DOI](#)). Mice were housed under a 12-h/12-h light-dark cycle, and experiments were performed during the dark cycle of mice. Mice were anesthetized with isoflurane (1.5 - 1.8%) and acepromazine (1.5 - 1.8 mg/kg body weight) when performing visual cortex craniotomy. Carprofen (5 mg/kg body weight) was administered prior to surgery. The body temperature was maintained using physically activated heat packs during surgery. Mouse eyes were kept moist with ointment during surgery. The scalp over-laying the right visual cortex was removed, and a custom head-fixing imaging chamber with a 5-mm diameter opening was mounted to the skull with cyanoacrylatebased glue (Oasis Medical) and dental acrylic (Lang Dental). A 4-mm diameter craniotomy was performed over the visual cortex and covered with #1 thickness coverslip.

Locating visual areas with intrinsic signal optical imaging (ISOI)

ISOI experiments were carried out similarly as previously (Stirman et al., 2016 [DOI](#); Smith et al., 2017 [DOI](#); Smith and Trachtenberg, 2007 [DOI](#)). Briefly, the pial vasculature images and intrinsic signal images were collected using CCD camera (Teledyne DALSA 1M30) at the craniotomy window. A $4.7 \times 4.7 \text{ mm}^2$ cortical area was imaged at $9.2 \text{ }\mu\text{m}/\text{pixel}$ spatial resolution and at 30 Hz frame rate. The pial vasculature was illuminated and captured through green filters ($550 \pm 50 \text{ nm}$ and $560 \pm 5 \text{ nm}$, Edmund Optics). The ISO image was collected by focusing $600 \text{ }\mu\text{m}$ down from the pial surface. The intrinsic signals were illuminated and captured through red filters ($700 \pm 38 \text{ nm}$, Chroma and $700 \pm 5 \text{ nm}$, Edmund Optics). Custom ISOI instruments were adapted from Kalatsky and Stryker (2003) [DOI](#). Custom acquisition software for ISOI imaging collection was adapted from David Ferster (Stirman et al., 2016 [DOI](#)). During ISOI, mice were 20 cm from a flat monitor ($60 \times 34 \text{ cm}^2$), which covers the visual field ($110^\circ \times 75^\circ$) of the left eye. Mice were lightly anesthetized with isoflurane (0.5%) and acepromazine (1.5–3 mg/kg). The body temperature was maintained at 37°C using a custom electric heat pad (Stirman et al., 2016 [DOI](#)). Intrinsic signal response to the vertical and horizontal drifting bar was used to generate azimuth and elevation retinotopic maps (Figure S1A [DOI](#)). The retinotopic maps were then used to locate V1 and HVAs. Borders between these areas

were drawn at the meridian of elevation and azimuth retinotopy manually (Marshall et al., 2011 [↗](#); Smith et al., 2017 [↗](#)). The vasculature map then provided landmarks to identify visual areas in two-photon imaging.

In vivo two-photon calcium imaging

Two-photon imaging was carried out using a custom Trepan2p microscope controlled by custom LabView software (Stirman et al., 2016 [↗](#)). Simultaneous dual-region imaging was achieved by splitting the excitation beam and temporally multiplexing laser pulses (Stirman et al., 2016 [↗](#)). Two-photon excitation light from an 80 MHz Ti:Sapph laser (Newport Spectra-Physics Mai Tai DeepSee) was split into two beams through polarization optics, and one path was delayed 6.25 ns relative to the other. The two beams were independently directed with custom voice-coil actuated steering mirrors and tunable lenses, such that the X, Y, Z planes of the two paths are independently positioned within the full field (4.4 mm diameter). Both beams were scanned by the resonant scanner (4 kHz, Cambridge Technologies), and a single photon signal was collected by a photomultiplier tube (PMT) (H7422P-40, Hamamatsu), and demultiplexed using outboard electronics prior to digitization. In the current study, two-photon imaging regions of $500 \times 500 \mu\text{m}^2$ were collected at 13.3 Hz for two-region imaging or 6.67 Hz for quad-region imaging. Imaging was performed with $< 80 \text{ mW}$ of excitation (910 nm) laser power, as measured out of the front of the objective. Mice recovered in their home cage for at least 2 days after surgery, before acquiring two-photon imaging. Mice were head-fixed $\sim 11 \text{ cm}$ from a flat monitor, with their left eye facing the monitor, during imaging. Approximately $70^\circ \times 45^\circ$ of the left visual field was covered. If not otherwise stated, two-photon images were recorded from quiet awake mice. For anesthetized experiments, mice were lightly anesthetized under 1% isoflurane.

In a subset of experiments, we monitored mouse pupil position and diameter using a custom-controlled CCD camera (GigE, National Instruments) at 20 - 25 fps. No additional light stimulation was used for pupil imaging. Pupil dynamics was analyzed using a custom-written imaging processing code in Matlab.

Visual stimuli

Visual stimulation was displayed on a 60 Hz LCD monitor ($9.2 \times 15 \text{ cm}^2$). All stimuli were displayed in full contrast. For coarse population RF and single neuron RF mapping (Figure S1B-D [↗](#)), a rectangular ($7.5^\circ \times 8.8^\circ$) bright moving patch containing vertical drifting grating (2 Hz, 0.05 cpd) on a dark background was displayed. The moving patch appeared and disappeared on a random position of the full monitor in pseudo-random order without interruption by a gray screen, and presented on each position for 5 sec.

To characterize the value and structure of the correlation of V1 and HVAs, we showed mice full-screen sinewave drifting grating stimuli in 8 directions ($0 - 315^\circ$, in 45° step), with an of 0.02, 0.05 or 0.19 cpd, and a TF of 1, 2 or 8 Hz (72 conditions in total). Each of the sine-wave drifting grating stimuli was presented for 2 s in pseudo-random order. Stimuli with the same SF and TF were presented successively without interruption. A gray screen was presented for 3 seconds when changing the SF and TF of stimuli. Each stimulus was presented 7-25 times, 15.2 trials on average.

In a subset of experiments, we also characterized the cross-stimulus stability of functional networks using combo stimuli with naturalistic videos and full contrast drifting gratings (at 2 Hz, 0.05 cpd). Two naturalistic videos, each lasting for 32 s were generated by navigating a mouse home cage using a GoPro camera. Each stimulus was presented for 10-20 times, 19.8 trials on average, interleaved with an 8 s gray screen.

Calcium imaging processing

Calcium imaging processing was carried out using custom MATLAB code (Yu et al., 2022 [DOI](#)). Two-photon calcium imaging was motion corrected using Suite2p sub-pixel registration module (Pachitariu et al., 2016 [DOI](#)). Neuron ROIs and cellular calcium traces were extracted from imaging stacks using custom code adapted from Suite2p modules (Pachitariu et al., 2016 [DOI](#)). Neuropil contamination was corrected by subtracting the common time series (1st principal component) of a spherical surrounding mask of each neuron from the cellular calcium traces (Harris et al., 2016 [DOI](#)). Neuropil contamination corrected calcium traces were then deconvolved using a Markov chain Monte Carlo (MCMC) method (Pnevmatikakis et al., 2013 [DOI](#)). For each calcium trace, we repeated the MCMC simulation for 400 times and measured the signal-to-noise of MCMC spike train inference for each cell. For all subsequent analyses, only cells that reliable spike train inference results were included. Neurons with low responsiveness were excluded for subsequent analysis (trial averaged spike count to preferred spatiotemporal frequency summed over all orientations < 1; or trial averaged spike count to a 32 s naturalistic video < 1).

Receptive field analysis

We mapped RFs by reverse correlation of neuronal responses with the locations of the moving patch of drifting grating stimulus. For population RF mapping, population neuronal responses of simultaneously recorded neurons from a $500 \times 500 \mu\text{m}^2$ imaging window were reverse correlated with the stimulus locations. We found the best fit elliptical 2D Gaussian profile for the intensity map (F) of the single neuron RF or the population neuron RF by minimizing the least square error.

$$F = A * \exp(-M(\frac{(x - x_0)^2}{2\sigma_x^2} + \frac{(y - y_0)^2}{2\sigma_y^2}))$$

The amplitude (A), rotation matrix (M), centroid (x_0, y_0), and spread (σ_x, σ_y) of the 2D Gaussian are found through the least square fit. The long and short axes of the 2D Gaussian of single-neuron RF are estimated by $\sqrt{2\ln 2} * \sigma$, which are the half-width halfmaximal of the Gaussian spread.

Similarly, population RF is characterized by the full-width half-maximal of the best-fit 2D Gaussian spread (average over the long and short axes), and the size of the 2D Gaussian profile above the half-maximal.

Gaussian mixture model

To characterize the tuning properties without an investigator bias, we used a data-driven approach and neurons were clustered using a Gaussian mixture model (GMM) based on the trial-averaged responses to the drifting gratings. Only reliable responsive neurons were included for GMM analysis (trial-to-trial Pearson correlation of the inferred spike trains > 0.08, spike trains were binned at 500 ms). Neuronal responses of the whole population pooled over all tested areas, were first denoised and reduced dimension by minimizing the prediction error of the trial-averaged response using the principle components (PC). 45 PCs were kept for population responses to the drifting gratings. We also tested a wide range number of PCs (20 – 70), and we found the tuning group clustering was not affected by the number of PCs. Neurons collected from different visual areas and different animals were pooled together in training GMM. GMMs were trained using MATLAB build function *fitgmdist* with a wide range number of clusters. A model of 65 classes was selected based on the Bayesian information criterion. We also examined models with smaller (20, 30, and 45) or even larger numbers of classes (75), the overall results held regardless of the number of GMM classes. **Figure S2A** [DOI](#) shows the t-SNE embedding of GMM classes and the direction selectivity of each class. The size of each GMM class is shown in **figure S1F** [DOI](#). t-SNE was performed using matlab built-in function *tsne*, with Mahalanobis distance method.

To summarize the spatial-temporal tuning properties of neurons (**Figure 1E**), we manually organized the 65 GMM classes into 6 groups based on their preferred SF-TF (**Figure S2A**). Group 1 prefers low SF and low TF (LSLT, 0.02 cpd, 1-2 Hz), group 2 prefers medium SF and low TF (MSLT, 0.05 cpd, 1-2 Hz), group 3 prefers high SF low TF (HSLT, 0.19 cpd, 1-2 Hz), group 4 prefers low SF high TF (LSHT, 0.02 cpd, 8 Hz), group 5 prefers medium SF and high TF (MSHT, 0.05 cpd, 8 Hz), group 6 prefers high SF and high TF (HSHT, 0.19 cpd, 8 Hz) and group 7 not specific (**Figure S2B**). Group 7 included 4 classes that did not exhibit specific response features, among them two classes are extremely small (each contains <5 neurons), and the other two contain neurons with small response strength (mean spike count < 0.5 spikes/s to preferred stimulus). As we have been inclusive in data selection for the GMM training and included low-firing neurons, the latter two classes contain about 1500 neurons. It is justifiable to exclude low-response neurons from further analysis. Thus, the whole group 7 was excluded for further analysis.

GMM classification accuracy

We examined the accuracy of GMM classification for neuron responses to drifting gratings. We performed GMM clustering on 10 random subsets of neurons (90% of all neurons) and measured the clustering consistency. First, we measured the consistency of the centers of the Gaussian clusters, which are 45D vectors in the PC dimensions. We measured the Pearson correlation of Gaussian center vectors independently defined by GMM clustering on random subsets of neurons. We found the center of the Gaussian profile of each class was consistent (**Figure 1C**). The same class of different GMMs was identified by matching the center of the class. Then, we asked whether a neuron was classified in the same class in each GMM model. We found neurons were consistently classified into the same class in GMMs of a random subset of data (**Figure 1D**). We also performed GMM on population data after randomly shuffling neuron identity (10 permutations). Classes were identified by matching the center of the class and then grouped following the previous definition. We found that neurons are allocated into the same SF-TF group in GMMs of randomly ordered data (**Figure S1G, H**). These analyses suggested that GMM provided a reliable classification of neurons.

Orientation and direction selectivity

The direction and orientation selectivity of each neuron was computed using the spatiotemporal frequency of drifting grating stimuli that drove the strongest response for that neuron. The direction selectivity index and orientation selectivity index were computed using the following equations.

$$SI = \frac{(R_{prefer} - R_{null})}{(R_{prefer} + R_{null})} \quad (1)$$

The polar plots of tuning groups were generated by averaging responses to the preferred direction of each neuron within a tuning group and normalized to one (**Figure S3C**). For neurons with high direction selectivity, neuron responses to the preferred direction were considered, while for neurons with low direction selectivity (DSI < 0.5), neuron responses to both preferred and null directions were included.

ISOI warping

We spatially registered the ISOI map of V1 to align with that of LM or AL (**Figure S6C-H**). We first segmented the ISOI map by color segmentation using K-means clustering and then determined the center of each color segment. Then we performed the affine transformation of color band centers of V1 to match that of LM or AL. The transformation matrix M was determined by minimizing the distance between transformed V1 centers and LM or AL centers using Matlab function *fminsearch*.

Correlation calculation

Signal correlation was defined as the correlation of stimulus tuning between neurons. The stimulus tuning of each neuron was estimated by the mean response of trials to the same stimulus ($\bar{Y}_i$), i is the neuron identity).

$$r_{sig} = \frac{cov(\bar{Y}_i, \bar{Y}_j)}{\sqrt{var(\bar{Y}_i) * var(\bar{Y}_j)}}$$

Noise correlation was defined as the trial-to-trial correlation of residual spike count (1 s time window, if not otherwise stated) after subtracting the mean response to each stimulus of the 72-condition sine-wave drifting gratings. Residual spike count to all stimuli (eg. gratings with different directions and SFs and TFs), and all trials were concatenated into one column vector per neuron (u_i , $u_i = Y_i - \bar{Y}_i$). The noise correlation r_{sc} was computed as the Pearson correlation of u_i and u_j .

$$r_{sc} = \frac{cov(u_i, u_j)}{\sqrt{var(u_i) * var(u_j)}}$$

i, j indicates neuron identity. Signal correlation was defined as the neuron-to-neuron Pearson correlation of mean responses. Mean response was a 72-element column vector, computed by trial averaging responses to sine-wave gratings with 72 conditions. To examine the relation between noise correlation and joint firing rate between a pair of neurons, we computed the mean joint spike count (geometric mean spike count average over all stimuli).

We computed inter-area NCs with simultaneously recorded regions that shared greater than 40% of population RF. We kept this criterion even though we did not detect a relationship between the inter-area NC and the fraction of population RF overlap within the tested range ($p = 0.37$).

Fidelity of noise correlation measurement

Tolerance of correlation calculation to inaccuracy in spike train inference

We quantify the spike train inference accuracy using a previously published data set with simultaneous cell-attached recording and two-photon imaging of GCaMP6s from mouse V1 (Chen et al. (2013) <http://cncs.org/data-sets/methods/cai-1>). We performed spike train inference on the recordings with stable baseline and good correspondence between calcium trace and electrophysiology recording (linear correlation, $r > 0.1$; bin 0.1 s; **Figure S4A, B**). The signal-to-noise (SNR) of the calcium trace of the calibration data is 12.3 ± 5 . It is comparable with the SNR of the calcium signal of the current study (8.7 ± 1.8).

We further evaluated how the correlation calculation was affected by inaccurate spikes train recovery. We took publicly available electrophysiology recordings of mouse V1 neurons (Theis et al. (2016) <http://spikefinder.codeneuro.org/>), and computed residual spike count correlation at 1 s time bin after perturbations on the ground truth spikes train. We did four types of perturbations, (1) randomly missing spikes; (2) missing isolated spikes as the signal-to-noise of the calcium signal of isolated spikes may be low; (3) missing all spikes within a burst; (4) missing 60% spikes within a burst (**Figure S4C**). We identified isolated spikes or burst spikes by thresholding the inter-spike-interval of each spike. A spike that was $> t$ s away from spikes flanking itself was a t

isolated spike. A spike that was $< t$ s away from another spike was a t burst spike. The residual spike count correlation computed with perturbed spikes trains was linearly correlated with ground truth (**Figure S4D**) and exhibited good tolerance to up to 60% missing spikes by all types of perturbation (fidelity > 0.6 ; **Figure S4E**).

Significance of noise correlation

Since the value of noise correlations was small, we tested whether these values were significantly above zero. We compared the noise correlation with trial-shuffled noise correlation, the latter was computed using trial-shuffled data (the order of trials were randomized for each neuron independently). The population-mean noise correlation computed with trial-aligned data was significantly higher than that of the trial-shuffled data with the size of the experimental population (**Figure 2B**).

Accuracy of noise correlation

We investigated the accuracy of noise correlation estimation with both data and model. The individual noise correlations of the same set of neurons varied when computing using a different random subset of trials (**Figure 2D**). We computed the population mean value of the noise correlation of a random subset of neuron pairs and calculated the confidence interval for estimating the population mean noise correlation. The accuracy of population-mean estimation increases with the number of neurons, even with a limited number of trials (**Figure 2C**). We further characterized the estimation accuracy by simulating correlated neuron population (Macke et al., 2009), which allows an arbitrary number of trials. The expected firing rate and expected population mean correlation match our experimental data. To achieve an accurate estimation (1/10 standard error/mean value) of the population mean correlation converges with >100 neurons even using experimental level trial numbers (**Figure 2C**).

Community module analysis

We constructed a V1-HVA connectivity matrix using the fraction of high NC ($NC > \text{mean} + 2.5 \times \text{SD}$ of trial-shuffled NC) pairs between each GMM class. We performed community detection analysis using the Louvain algorithm (Rubinov and Sporns, 2010), which assigned densely connected nodes to the same module. The spatial smooth parameter γ that generated the largest deviation from a random connectivity matrix is picked. The analysis was performed using the Brain Connectivity Toolbox (*brain-connectivity-toolbox.net*).

Leaky integrate-and-fire neuron network simulation

LIF simulations were carried out using the Brian2 simulation engine in Python (Stimberg et al., 2019). The LIF neuron network model was defined similarly as Song et al. (2000). In brief, the membrane potential of LIF neurons was given by the equation below:

$$\frac{dv}{dt} = (ge * (Ee - v) + gi * (Ei - v) + El - v) / \tau_m$$

Where τ_m corresponds to the membrane time constant (20 ms). ge , gi and Ee , Ei are the excitatory and inhibitory synaptic conductance and their respective reversal potential ($Ee = 0$ mV, $Ei = -80$ mV). The membrane potential was simulated with a time resolution of $dt = 0.1$ ms. El (-70 mV) corresponds to the resting potential. The dynamics of synaptic conductance were given by exponential decay functions $ge/dt = -ge/\tau_e$ and $gi/dt = -gi/\tau_i$ where τ_e (5 ms) and τ_i (10 ms) are the decay time constants for excitatory and inhibitory synapses.

Synaptic connections between LIF neurons occurred with probability $p = 0.02$, and the strength of the connections is defined as W_{ij}^{rec} , (i, j indicate source and target LIF neuron ID, $i \neq j$).

$W_{ij}^{rec} = J_{max}$ or 0 if not connected. In simulations for [figure 6D](#) J_{max} ranges from 0.05 – 0.3, while in simulations for [figure 6E-H](#) J_{max} is fixed to 0.2 for closed markers or 0 for open markers.

The LIF network received feedforward (FFD) stimulus input from Poisson neurons ($N = 1000$ in the network simulation, and $N = 80$ in the toy model), whose firing follows time-varying Poisson processes (0 – 30 Hz; [Figure 6C](#) insert). The instant firing rate of the FFD Poisson neuron is defined by combining five weighted Gaussian profiles. The firing rate of the FFD input neuron varies for different types of stimuli by randomly initializing the weights to the five Gaussian profiles. In simulations with top-down input, we adjusted the FFD firing rate by changing the amplitude of the Gaussian profile for moderate or strong top-down input while keeping the LIF network firing rate to 4 – 6 Hz.

The Poisson FFD input neurons are connected to LIF neurons with a probability of $p = 0.2$, and the strength of the connections is defined as W_{lj}^{ff} . $W_{lj}^{ff} = S_{lj}$ or 0 if not connected. l indicates the source FFD input neuron.

$$S_{l,j} = a * 0.2 * \exp(-((i/10.0 - j)/b)^2) + c * 0.2$$

a , b , and c are parameters that manipulate the structure of the FFD connection, ranging from fully random to fully bell-shaped ([Figure 6E](#)). FFD connection set to random in simulations for [figure 6D, G, H](#).

Top-down inputs in [figure 6G-H](#) are generated by d independent Poisson process. d defines the dimensionality of the top-down input, ranging from 1–100 for [figure 6G](#) and fixed $d = 10$ for [figure 6H](#). The instant firing rate of these neurons is pooled from a uniform random distribution and changes on each trial. The firing rate of these neurons ranges from 1 – 150 Hz for generating moderate to strong top-down input to the LIF neurons. The top-down input neurons are randomly connected with the LIF network with a probability of $p = 0.2$. The strength of the top-down connections is defined as W_{kj}^{td} . $W_{kj}^{td} = G_{kj}$ or 0 if not connected. G_{kj} is generated by a uniform distribution between 0 – 0.2. k indicates the source top-down input neuron. The stimulus/top-down input ratio is determined by the changes in membrane potential induced by each source of input.

In the toy model, the connectivity S_{ij} equals the fraction of shared input. The Poisson input neuron fires constantly at 5 Hz.

Decompose neural activity

We use linear fitting to estimate the fraction of population neural activity explained by stimulus input, pupil dynamics, and network interactions ([Figure S8B](#)). First, we estimated the variance of population neural activity due to stimulus input. The expected stimulus-response of each neuron is estimated by the average response of trials to the same stimulus. The contribution of the stimulus-driven portion of population neural activity is defined by the equation below, or the variance of the expected stimulus response divided by the total variance (i is stimulus ID; t is trial ID).

$$Frac_{stim} = Var(\bar{Y}_i) / Var(Y_{it})$$

After subtracting the stimulus-driven portion of population neural activity ($Y_{res} = Y_{it} - \bar{Y}_i$), we next measure the contribution from pupil dynamics (X_{pupil}) to population neural activity using a reduced rank regression model. The relationship between residual population neural activity and pupil dynamics is defined as

$$Y_{res} = X_{pupil} * B$$

The reduced rank objective can be written as

$$\min_B tr[(Y_{res} - X_{pupil}B)(Y_{res} - X_{pupil}B)^T] \\ s.t. rank(B) \leq r$$

The coefficient of the reduced rank regression model B_c is estimated by

$$B_c = B_{OLS} * VV^T$$

B_{OLS} is the ordinary least squares solution, and $\hat{Y}_{OLS} = X_{pupil} * B_{OLS}$. V is the first r eigenvectors of $\hat{Y}_{OLS}^T \hat{Y}_{OLS}$.

In this model, the pupil dynamics matrix is composed of pupil area, centroid position, and principal components of the pupil video that explain more than 1% of the variance of the video (**Figure S8A**), together they composed the factor matrix of B . On average 11.2 principle components were kept and accounted for 87% of the variance of the video. Both the reduced rank regression model and the ordinary regression model generated quantitatively similar results. However, a reduced rank regression model is recommended to constrain the fitting better and avoid rank deficiency problems. The rank dimension is 13.4 ± 3 . With the solution of reduced rank regression (B_c) and prediction ($\hat{Y}_{RRR} = X_{pupil} * B_c$), we estimated the contribution of pupil dynamics to neural activity.

$$Frac_{pupil} = Var(\hat{Y}_{RRR}) / Var(Y_{it})$$

Next, we measured the contribution of network interactions to the unexplained portion of neural activities, which was computed by subtracting the pupil-related firing from the residual neural activity ($Y_{unexp} = Y_{res} - \hat{Y}_{RRR}$). We model the unexplained portion of individual neural activity ($Y_{unexp}^{(n)}$ for target neuron n) with the rest of the network using a principal component regression model. Principle components of unexplained neural activity (exclude one target neuron at a time, $PC_{Score}^{(-n)}$) with > 0.1 variance were kept.

$$Y_{unexp}^{(n)} = PC_{Score}^{(-n)} * B_{pc}$$

The coefficients of the equation were estimated using an ordinary least square solution. We estimated the neural activity that was generated by network interaction as

$\hat{Y}_{net} = PC_{Score}^{(-n)} * B_{pc}$. The contribution of network interaction to neural activity is measured by

$$Frac_{net} = Var(\hat{Y}_{net}) / Var(Y_{it})$$

Code and data availability

Code and data are available upon request.

Supplementary Information

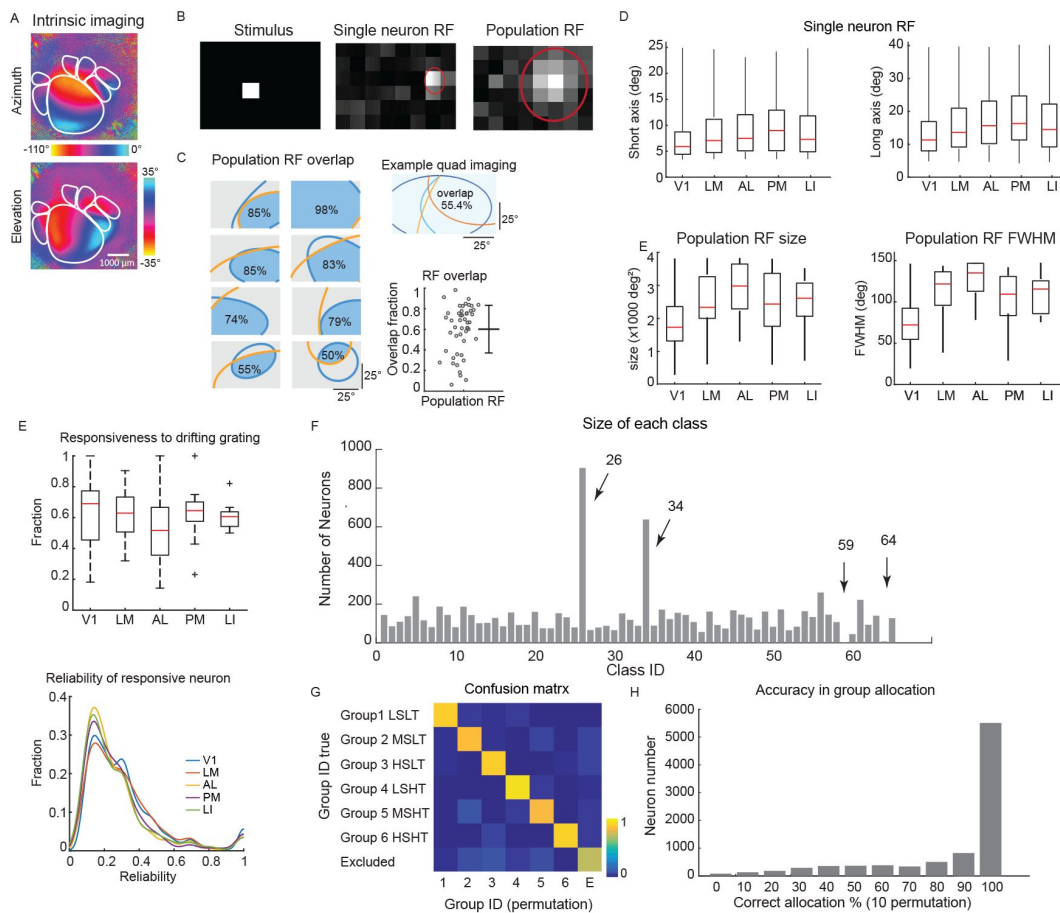


Figure S1.

Functional groups by multi-region two-photon calcium imaging.

(A) Example intrinsic signal imaging of mouse visual areas. (B) Moving square stimuli for quick RF mapping. (C) Example population RFs of simultaneously imaged populations. Blue and orange contours indicate the Gaussian profile of population RF of neurons from different visual areas, and blue shade indicates the overlap region of population RF of two simultaneous imaging regions. Values indicate the fraction of overlap. Upper right: example population RF of a quartic-region imaging. Lower right: summarize the fraction of population RF overlap of individual experiments (gray circle). Error bars indicate the mean and standard deviation. (D) Upper: short and long axes of the Gaussian profile of single neuron RF of all tested HVA neurons are longer than that of V1 (short, $p < 0.0001$; long, $p < 0.0001$; One-way ANOVA with Bonferroni correction). Bottom: population RFs of HVA are significantly larger than that of V1 (FWHM, $p = 0.0003$; Size: $p < 0.0001$. one-way ANOVA with Bonferroni correction). (E) The responsiveness of V1 and HVAs to the 72-condition sine-wave drifting grating stimuli. Left: the fraction of responsive neurons in HVAs is not significantly different (trial-to-trial Pearson correlation > 0.08 ; one-way ANOVA, $p = 0.36$). Right: distribution of neuron firing reliability (trial-to-trial Pearson correlation of inferred spike train at 500 ms bin). Only responsive neuron was considered. V1 and LM were slightly more reliable than AL, PM, and LI (one-way ANOVA with Bonferroni multiple comparisons, $p = 1.7 \times 10^{-7}$). (F) Number of neurons of each GMM class. Classes 26 and 34 were left unclassified, as they were largely insensitive to stimulus parameters, and classes 59 and 64 were small and thus excluded from further analysis. (G) The confusion matrix shows the joint probability of a neuron being identified as group A in GMM of randomly ordered data (shuffle the order of neurons, horizontal), and the neuron is classified as group B in GMM of the original data set (vertical). The diagonal indicates the probability of neurons being classified into the same group. The confusion matrix was generated by averaging a joint probability of 10 permutations. (H) The bar chart shows the probability of correctly allocating neurons into the same group in 10 permutations of GMMs of randomly ordered data.

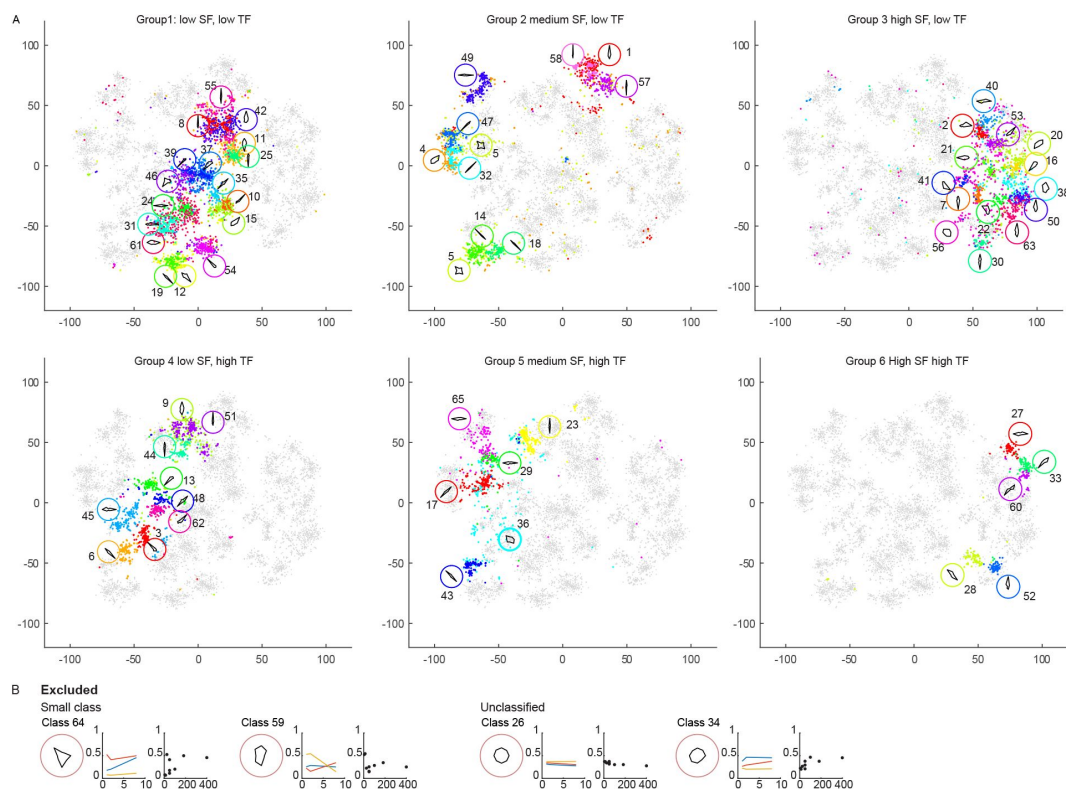


Figure S2.

t-SNE embedding of GMM classes.

(A) t-SNE embedding of all the neurons for the SF-TF analysis. Axes are arbitrary units. Neurons are colored by their GMM class ID (inserted number) and are organized into 6 groups by SF-TF preference. A polar plot shows the average tuning curve for the class is shown in the same color. (B) Four GMM classes are excluded due to being too small or having no characteristic feature selectivity. The response feature of each class is described by three panels: a polar plot shows the average tuning curve for cells in the class; the middle panel shows the normalized response to different joint combinations of TF (x-axis, Hz) and SF (colored line, blue 0.02 cpd, red 0.05 cpd, yellow 0.19 cpd); right panel shows the normalized response to different speed of gratings (x-axis, deg/s).

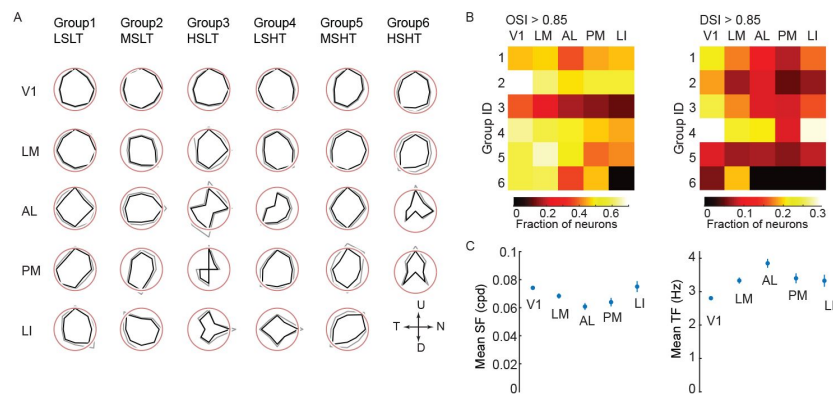


Figure S3.

Spatial modulation on SF-TF and orientation tuning.

(A) Polar plots of averaged preferred directions of six tuning groups of V1 and HVAs. Polar plots were generated with >30 neurons. Black and gray lines indicate the mean and SEM of normalized preferred directions. (B) The heatmap shows the fraction of orientation (OSI > 0.85, left) and direction-selective neurons (DSI > 0.85, right) of each tuning group in V1, LM, AL, and PM. (C) Mean preferred SF (left) and TF (right) of tested visual areas. AL prefers lower SF and higher TF than the other tested visual areas ($p < 0.0001$, one-way ANOVA with Bonferroni multi-comparison).

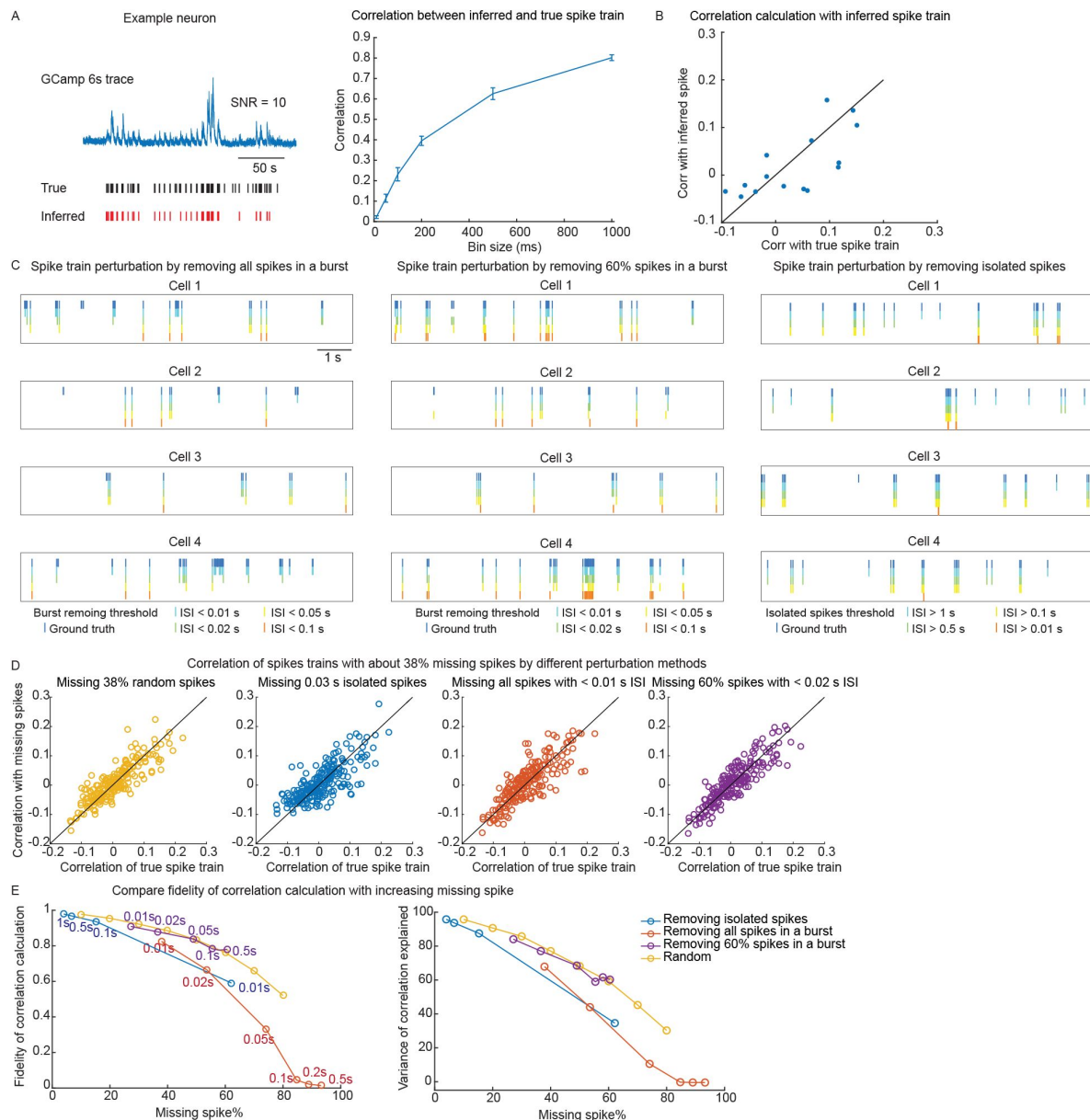


Figure S4.

Tolerance of noise correlation to missing spikes.

(A) Left, comparing the inferred spike train and ground truth spike train (cell-attached recording) of one example neuron. Spike inference recovered 50% of the spikes of this neuron, the linear correlation between the inferred spike train and the true spike train is 0.79 (bin 1 s). Right, the correlation between the inferred spike train and the true spike train at various time bins. (B) The inter-neuron cross-correlation computed by true spike train and inferred spike are linearly correlated ($r = 0.7$). (C) The ground truth spike trains (top) and spike train after different types of perturbations of example neurons from the spikefinder dataset (Methods). (D) Compute correlation of residual spike count at 1 s time bin after spike perturbations from left to right: random missing spikes; missing isolated spikes with inter-spike-interval (ISI) > 0.03 s; missing all spikes within a burst ISI < 0.01 s; missing 60% spikes within a burst with ISI < 0.02 s. (E) Fidelity (left) and variance explained (right) of correlation calculation with spike train perturbation. The fidelity was defined as the linear correlation between spike count correlation before and after perturbation. Variance explained was measured as r^2 of a linear regression between true correlation and perturbed correlations. The colored text in the figure indicates the ISI thresholds.

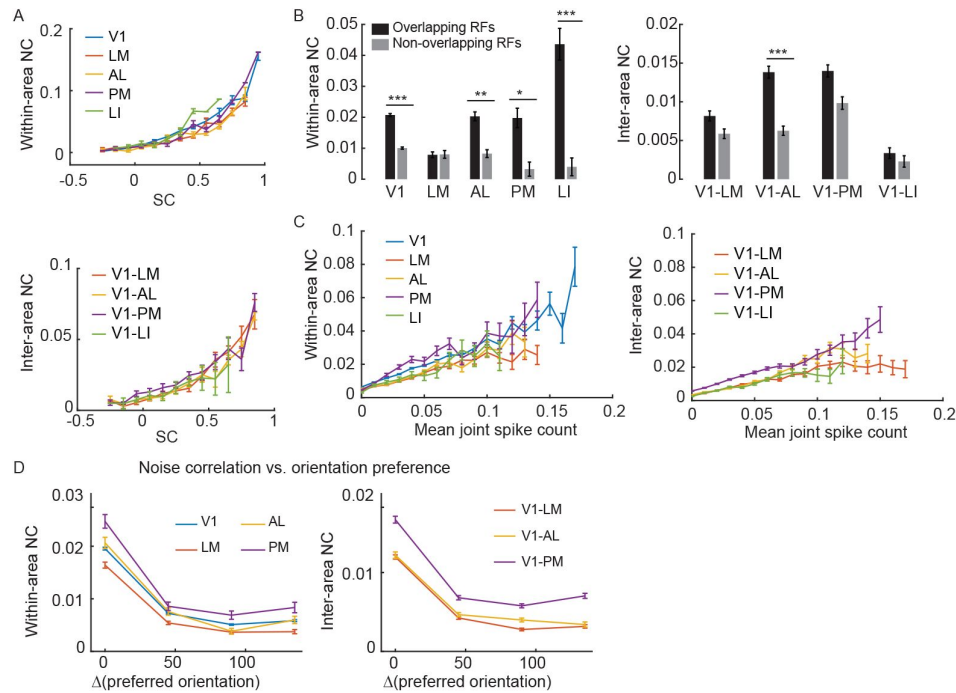


Figure S5.

Factors contribute to the variance of NCs.

(A) Within- and inter-area noise correlations are positively related to signal correlation. (B) Within-area (left) and inter-area (right) NC is significantly higher in neuron pairs with shared RF (within-area, $PV1 < 0.0001$, $PLM = 0.98$, $PAL = 0.006$, $PPM = 0.03$, $PLI < 0.0001$; inter-area: $PV1-LM = 0.3$, $PAL = 0.0007$, $PV1-PM = 0.15$, $PV1-LI = 0.82$; t-test). Overlapping groups and non-overlapping groups are defined as neuron pairs share $> 60\%$ RF, and $< 20\%$ RF, respectively. (C) Noise correlations of V1 and HVAs are positively related to joint spike count (For all within- and inter-area correlation, $r = 0.09-0.18$, $p < 0.0001$). Mean joint spike count is the geometric mean of the spike count to all stimuli. (D) Plot within- and inter-area noise correlation as a function of difference in preferred orientation. Only orientation-selective neurons ($OSI > 0.5$) were included.

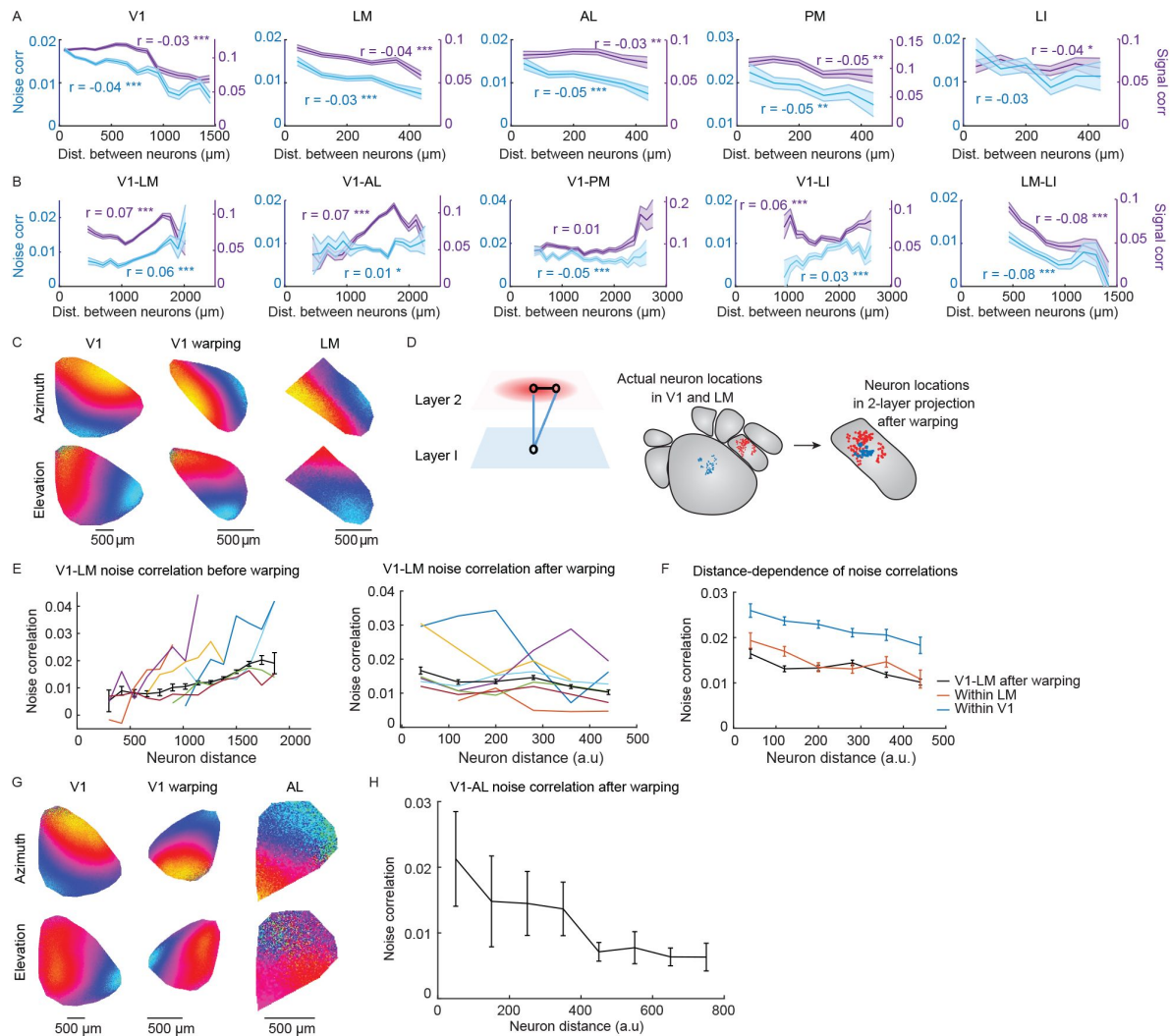


Figure S6.

Distance-dependence of inter-area NC explained by retinotopic map.

(A) The distance-dependent decrease of within-area NCs (blue) was not different among V1, LM, AL, and PM with a linear decreasing rate of NC = $-0.02 \pm 0.009 \text{ mm}^{-1}$ (mean $\pm$ S.D.). In these areas, SCs (purple) also exhibited significant distance-dependent decreases. (B) Inter-area NC (blue) and SC (purple) exhibited region-specific distance-dependence patterns that were often non-monotonic. (A-B) Solid lines and shaded areas indicate the mean and standard error of the mean. Each distance bin contains >50 data points. Pearson correlation r is shown in the figure and stars indicate significance (*, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.0001$). (C) Example affine transformation of ISOI maps. The left shows the original V1 map, the middle is the V1 map after affine transformation and the right is the original LM map. (D) Left, is a cartoon of two recurrent layers with an aligned retinotopic map. Right: neuron location on the visual cortex before and after warping. To examine whether the distance-dependent increase of V1-LM NCs to drifting gratings could be explained by retinotopy, we aligned V1 and LM retinotopy by affine transformations of intrinsic signal retinotopic maps of V1 to match those of LM. After retinotopic warping, V1 and LM were treated as two layers with aligned retinotopies (E) Distance-dependent increasing of V1-LM NC to sine-wave drifting gratings before (left) and after retinotopic warping (right). Individual experiments with significant distance dependence are shown in colored curves. The black curve shows the population mean and standard error (Pearson correlation, before warping, $r = 0.066$, $p < 0.0001$; after warping, $r = -0.026$, $p < 0.0001$). (F) After warping, the inter-area NCs between V1 and LM exhibited a distance-dependent decrease (linear correlation, $r = -0.26$, $p = 3 \times 10^{-6}$), similar to the trends found for within-area NCs in V1 and LM. (G) Example affine transformation of ISOI maps of V1 and AL. The left shows the original V1 map, the middle is the V1 map after affine transformation and the right is the original AL map. (H) Distance-dependent decreasing of inter V1-AL NC after retinotopic alignment (linear correlation, $r = -0.05$, $p < 0.0001$).

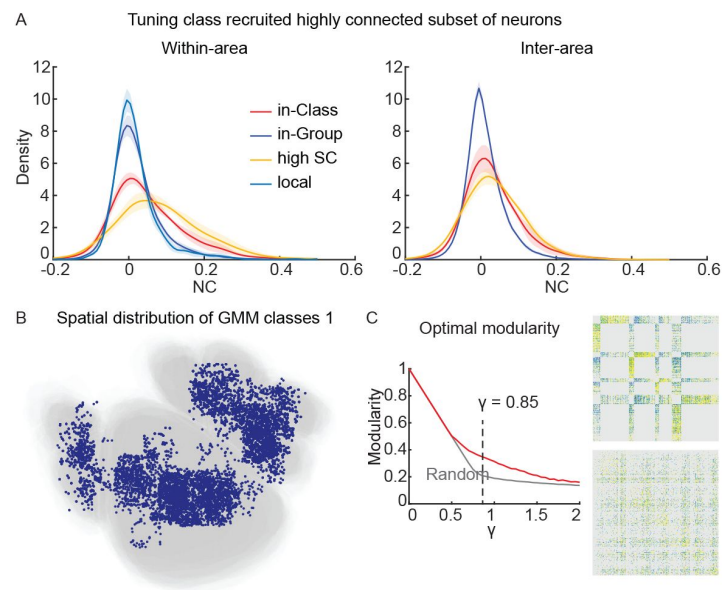


Figure S7.

Connectivity between GMM classes

(**A**) The density function of within-area (left) and inter-area (right) NC for neuron pairs from the same tuning group, or the same GMM class, shared high SC, or from the local neighborhood. The SC or the distance threshold is defined such that the same number of neuron pairs is included as the in-class neuron pairs. (**B**) The spread of neurons from GMM class 1 on a registered map for visual areas. (**C**) The modularity of the V1-HVA connectivity between GMM classes is regulated by a spatial smooth parameter γ . We reported the modular structure of the connectivity matrix when $\gamma = 0.85$, as it generated the largest deviation from a random connectivity matrix. The left side shows the modular structure of V1-HVA GMM classes (upper), and that of a random matrix preserving the degree distribution (lower).

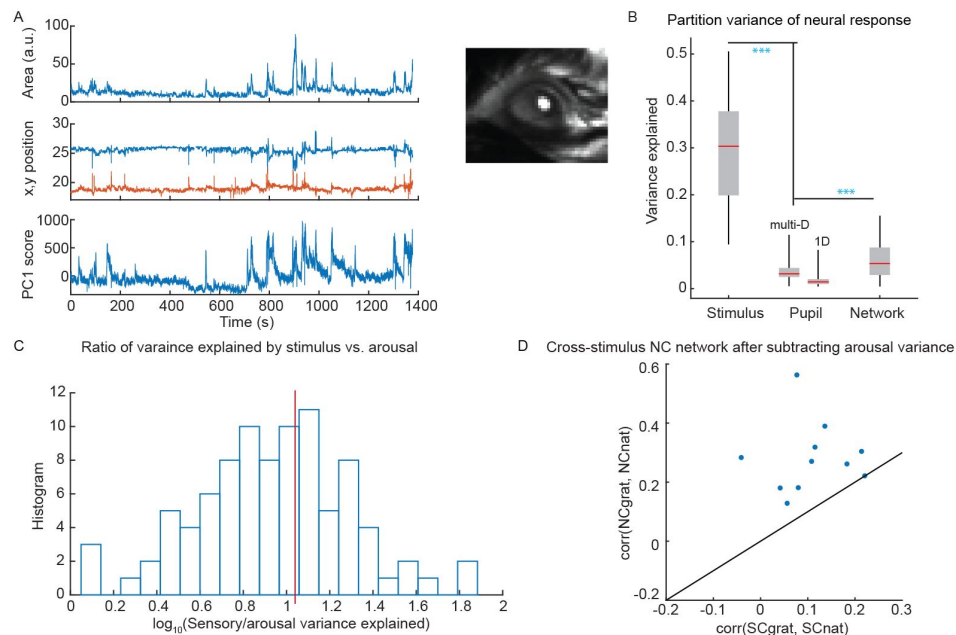


Figure S8.

Gain modulation does not explain the NC connectivity

(A) Example time series of pupil dynamics: the pupil area (pixel), x,y position of the pupil centroid, and the first PC of the pupil video. (B) Decomposition of the variance of population neural activity to stimulus, pupil dynamics, and network interaction. Pupil dynamics explained less variance than stimulus and network interactions (***, $p < 0.0001$, paired t-test). A multi-dimensional pupil dynamics model explains more variance than a 1D model (Multi-D $3.5 \pm 2\%$, 1D $1.5 \pm 1\%$, $p < 0.0001$, paired t-test). The rank of the multi-dimensional model is the best-fit model of reduced rank regression, the rank dimension is 13.4 ± 3 . The 1D model is generated by the most pronounced eigenvector of the reduced rank regression model. (C) To examine the relative contributions of stimulus and pupil dynamics, we analyzed each experiment and plotted a histogram of the ratio of neural activity variance explained by stimulus input and pupil dynamics (arousal modulation). Note that the x-axis is in $\log_{10}$ scale. The red line indicates the mean of the distribution, which indicates that the stimulus modulates neuronal activity approximately 10-fold more than pupil dynamics / arousal. The analysis was only performed on the subset of experiments with pupil recordings (16 mice, 34 recording sessions, total of 91 cortical areas). (D) We reproduced the qualitative results in Figure 5A using only the residual neural activity after subtracting the arousal-modulated portion. The analysis was only performed on the subset of experiments with pupil recordings.

Supplementary Table

Gratings stimuli								
Animal ID	Recording areas				Total neurons per area			NC pairs
421	LM		AL		147		58	20910
421	LM		LM		93		19	6216
493	V1		V1		48		21	2346
382	V1		LM		208		37	29890
490	V1		LM		68		110	15576
490	V1		AL		63		69	8515
493	V1		AL		216		62	38226
493	V1		LM		159		104	34191
426	V1		AL		85		69	11781
426	V1		LM		47		39	3655
426	V1		PM		71		4	2628
382	V1		LM		82		111	18336
382	LM		LI		61		84	10296
382	V1		AL		35		44	3003
470	V1		AL		77		34	5778
470	V1		V1		36		24	1711
471	V1		PM		81		17	4753
471	V1		LM		20		29	1128
471	V1		V1		105		40	10440
391	V1		PM		51		22	2628
391	V1		V1		33		26	1711
417	V1		AL		84		40	7626
417	V1		LM		59		9	2278
400	V1		PM		11		7	153
400	V1		AL		18		7	300
532	V1		PM		302		52	62128
540	V1		AL		17		13	435
557	V1		LI		199		90	41616
559	V1		LM		18		18	630
577	V1		PM		21		41	1891
459	V1	AL	LM		24	116	25	13530
513	V1	LM	LI		40	90	43	14878
557	V1	PM	AL		283	69	18	68265
560	V1	V1	PM		100	127	28	32385
576	V1	V1	PM		38	86	88	22366
577	V1	AL	LM		117	123	11	31375
459	V1	PM	V1	AL	286	13	57	89676
525	V1	PM	AL	LM	214	204	187	256686
533	V1	AL	LM	LI	157	91	74	67528
540	V1	V1	LM	LI	38	39	9	3916
578	V1	V1	PM	AM	130	56	24	33670
560	V1	V1	PM	PM	217	305	74	234955
560	V1	V1	LM	AL	359	273	92	416328
540	V1	AL	LM	LI	113	163	61	79401
559	V1	LM	LM	LI	315	262	203	326836
632	V1	V1	LM	LI	290	306	139	343206
633	V1	V1	LM	LI	197	243	220	311655

Grating and natural video combo stimuli					
Animal ID	Recording areas		Total neurons per area		NC pairs (response to both stimuli)
143	V1	LM	121	57	4005
143	V1	LM	116	38	2415
143	V1	LM	66	47	666
144	V1	LM	119	3	1652
144	V1	PM	53	65	666
154	V1	AM/PM	247	118	3486
154	V1	AM	145	107	3160
154	V1	AM/PM	121	164	903
154	V1	V1	135	134	8256
156	V1	LM	129	87	7021
156	V1	AL	171	38	6903
166	V1	AL	304	162	10440
167	V1	AL	181	117	5565
170	V1	AL	352	163	10585
171	V1	AL	119	19	1830
171	V1	V1	161	123	5671
211	V1	V1	100	169	3486
633	V1	LM	85	150	2415
633	V1	V1	144	100	6441
635	V1	LM	94	141	2211
657	V1	LM	37	47	378
657	V1	LM	37	79	276
635	V1	AL	400	275	32385
635	V1	V1	287	343	36315

Combo	Total neurons	Total animals	Total within-area NC pair (response to both stimuli)
V1:	4573	12	45203
LM:	647	6	22850
AL:	774	6	6963
'AM/PM'	454	2	871

Gratings	Total neurons	Total animals	Total within-area NC pair
V1:	6735	23	675569
LM:	2076	17	143581
AL:	1352	14	79173
LI:	593	8	25148
'AM/PM'	786	12	38343

Figure T1.

Entire data list.

The Animal ID is a simple identifier number. Note that some animals were used for multiple imaging configurations. On the left is the information for mice that were imaged during viewing of the drifting grating visual stimuli only. On the right is the information for the mice that were imaged both during viewing of gratings and during viewing of the naturalistic video stimuli. On the bottom right are summary figures for the total numbers of animals, neurons, and unique neuron pairs (imaged simultaneously to permit the computation of noise correlations).

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Editors

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Reviewer #1 (Public review):

Summary:

Using multi-region two-photon calcium imaging, the manuscript meticulously explores the structure of noise correlations (NCs) across mouse visual cortex and uses this information to make inferences about the organization of communication channels between primary visual cortex (V1) and higher visual areas (HVAs). Using visual responses to grating stimuli, the manuscript identifies 6 tuning groups of visual cortex neurons, and finds that NCs are highest among neurons belonging to the same tuning group whether or not they are found in the same cortical area. The NCs depend on the similarity of tuning of the neurons (their signal correlations) but are preserved across different stimulus sets - noise correlations recorded using drifting gratings are highly correlated with those measured using naturalistic videos. Based on these findings, the manuscript concludes that populations of neurons with high NCs constitute discrete communication channels that convey visual signals within and across cortical areas.

Strengths:

Experiments and analyses are conducted to a high standard and the robustness of noise correlation measurements is carefully validated. To control for potential influences of behaviour-related top-down modulation of noise correlations, the manuscript uses measurements of pupil dynamics as a proxy for behavioural state and shows that this top-down modulation cannot explain the stability of noise correlations across stimuli.

Weaknesses:

The interpretation of noise correlation measurements as a proxy from network connectivity is fraught with challenges. While the data clearly indicate the existence of distributed functional ensembles, the notion of communication channels implies the existence of direct anatomical connections between them, which noise correlations cannot measure.

The traditional view of noise correlations is that they reflect direct connectivity or shared inputs between neurons. While it is valid in a broad sense, noise correlations may reflect shared top-down input as well as local or feedforward connectivity. This is particularly important since mouse cortical neurons are strongly modulated by spontaneous behavior (e.g. Stringer et al, Science, 2019). Therefore, noise correlation between a pair of neurons may reflect whether they are similarly modulated by behavioral state and overt spontaneous behaviors. Consequently, noise correlation alone cannot determine whether neurons belong to discrete communication channels.

<https://doi.org/10.7554/eLife.97848.2.sa3>

Reviewer #2 (Public review):

Summary:

This groundbreaking study characterizes the structure of activity correlations over millimeter scale in the mouse cortex with the goal of identifying visual channels, specialized conduits of visual information that show preferential connectivity. Examining the statistical structure of visual activity of L2/3 neurons, the study finds pairs of neurons located near each other or across distances of hundreds of micrometers with significantly correlated activity in response to visual stimuli. These highly correlated pairs have closely related visual tuning sharing orientation and/or spatial and/or temporal preference as would be expected from dedicated visual channels with specific connectivity.

Strengths:

The study presents best-in-class mesoscopic-scale 2-photon recordings from neuronal populations in pairs of visual areas (V1-LM, V1-PM, V1-AL, V1-LI). The study employs diverse visual stimuli that capture some of the specialization and heterogeneity of neuronal tuning in mouse visual areas. The rigorous data quantification takes into consideration functional cell groups as well as other variables that influence trial-to-trial correlations (similarity of tuning, neuronal distance, receptive field overlap, behavioral state). The paper demonstrates the robustness of the activity clustering analysis and of the activity correlation measurements. The paper shows convincingly that the correlation structure observed with grating stimuli is present in the responses to naturalistic stimuli. A simple simulation is provided that suggest that recurrent connectivity is required for the stimulus invariance of the results. The paper is well written and conceptually clear. The figures are beautiful and clear. The arguments are well laid out and the claims appear in large part supported by the data and analysis results (but see weaknesses).

Weaknesses:

An inherent limitation of the approach is that it cannot reveal which anatomical connectivity patterns are responsible for observed network structure. A methodological issue that does not seem completely addressed is whether the calcium imaging measurements with their limited sensitivity amplify the apparent dependence of noise correlations on the similarity of tuning. Although the paper shows that noise correlation measurements are robust to changes in firing rates / missing spikes, the effects of receptive field tuning dissimilarity are not addressed directly. The calcium responses of mouse visual cortical neurons are sharply tuned. Neurons with dissimilar receptive fields may show too little overlap in their estimated firing rates to infer noise correlations, which could lead to underestimation of correlations across groups of dissimilar neurons.

<https://doi.org/10.7554/eLife.97848.2.sa2>

Reviewer #3 (Public review):

Summary:

Yu et al harness the capabilities of mesoscopic 2P imaging to record simultaneously from populations of neurons in several visual cortical areas and measure their correlated variability. They first divide neurons in 65 classes depending on their tuning to moving gratings. They found the pairs of neurons of the same tuning class show higher noise correlations (NCs) both within and across cortical areas. Based on these observations and a model they conclude that visual information is broadcast across areas through multiple, discrete channels with little mixing across them.

NCs can reflect indirect or direct connectivity, or shared afferents between pairs of neurons, potentially providing insight on network organization. While NCs have been comprehensively studied in neurons pairs of the same area, the structure of these correlations across areas is much less known. Thus, the manuscripts present novel insights on the correlation structure of visual responses across multiple areas.

Strengths:

The measurements of shared variability across multiple areas are novel. The results are mostly well presented and many thorough controls for some metrics are included.

Weaknesses:

I have concerns that the observed large intra class/group NCs might not reflect connectivity but shared behaviorally driven multiplicative gain modulations of sensory evoked responses. In this case, the NC structure might not be due to the presence of discrete, multiple channels broadcasting visual information as concluded. I also find that the claim of multiple discrete broadcasting channels needs more support before discarding the alternative hypothesis that a continuum of tuning similarity explains the large NCs observed in groups of neurons.

Specifically:

Major concerns:

(1) Multiplicative gain modulation underlying correlated noise between similarly tuned neurons

(1a) The conclusion that visual information is broadcasted in discrete channels across visual areas relies on interpreting NC as reflecting, direct or indirect connectivity between pairs, or common inputs. However, a large fraction of the activity in the mouse visual system is known to reflect spontaneous and instructed movements, including locomotion and face movements, among others. Running activity and face movements are one of the largest contributors to

visual cortex activity and exert a multiplicative gain on sensory evoked responses (Niell et al, Stringer et al, among others). Thus, trial-by-fluctuations of behavioral state would result in gain modulations that, due to their multiplicative nature, would result in more shared variability in cotuned neurons, as multiplication affects neurons that are responding to the stimulus over those that are not responding (see Lin et al, Neuron 2015 for a similar point).

In the new version of the manuscript, behavioral modulations are explicitly considered in Figure S8. New analyses show that most of the variance of the neuronal responses is driven by the stimulus, rather than by behavioural variable. However, they new analyses still do not address if the shared noise correlation in cotuned neurons is also independent of behavioral modulations .

As behavioral modulations are not considered this confound affects the conclusions and the conclusion that activity in communicated unmixed across areas (results in Figure 4), as it would result in larger NCs the more similar the tuning of the neurons is, independently of any connectivity feature. It seems that this alternative hypothesis can explain the results without the need of discrete broadcasting channels or any particular network architecture and should be addressed to support the main claims.

(2) Discrete vs continuous communication channels

(2a) One of the author's main claims is that the mouse cortical network consists of discrete communication channels, as stated in the title of the paper. This discreteness is based on an unbiased clustering approach on the tuning of neurons, followed by a manual grouping into six categories with relation to the stimulus space. I believe there are several problems with this claim. First, this clustering approach is inherently trying to group neurons and discretise neural populations. To make the claim that there are 'discrete communication channels' the null hypothesis should be a continuous model. An explicit test in favor of a discrete model is lacking, i.e. are the results better explained using discrete groups vs. when considering only tuning similarity? Second, the fact that 65 classes are recovered (out of 72 conditions) and that manual clustering is necessary to arrive at the six categories is far from convincing that we need to think about categorically different subsets of neurons. That we should think of discrete communication channels is especially surprising in this context as the relevant stimulus parameter axes seem inherently continuous: spatial and temporal frequency. It is hard to motivate the biological need for a discretely organized cortical network to process these continuous input spaces.

Finally, as stated in point 1, the larger NCs observed within groups than across groups might be due to the multiplicative gain of state modulations, due to the larger tuning similarity of the neurons within a class or group.

<https://doi.org/10.7554/eLife.97848.2.sa1>

Author response:

The following is the authors' response to the original reviews.

General Response

We are grateful for the constructive comments from reviewers and the editor.

The main point converged on a potential alternative interpretation that top-down modulation to the visual cortex may be contributing to the NC connectivity we observed. For this revision, we address that point with new analysis in Fig. S8 and Fig. 6. These results indicate that top-down modulation does not account for the observed NC connectivity.

We performed the following analyses.

(1) In a subset of experiments, we recorded pupil dynamics while the mice were engaged in a passive visual stimulation experiment (Fig. S8A). We found that pupil dynamics, which indicate the arousal state of the animal, explained only 3% of the variance of neural dynamics. This is significantly smaller than the contribution of sensory stimuli and the activity of the surrounding neuronal population (Fig. S8B). In particular, the visual stimulus itself typically accounted for 10-fold more variance than pupil dynamics (Fig. S8C). This suggests that the population neural activity is highly stimulus-driven and that a large portion of functional connectivity is independent of top-down modulation. In addition, after subtracting the neural activity from the pupil-modulated portion, the cross-stimulus stability of the NC was preserved (Fig. S8D).

We note that the contribution from pupil dynamics to neural activity in this study is smaller than what was observed in an earlier study (Stringer et al. 2019 Science). That can be because mice were in quiet wakefulness in the current study, while mice were in spontaneous locomotion in the earlier study. We discuss this discrepancy in the main text, in the subsection “*Functional connectivity is not explained by the arousal state*”.

(2) We performed network simulations with top-down input (Fig. 6F-H). With multidimensional top-down input comparable to the experimental data, recurrent connections within the network are necessary to generate cross-stimulus stable NC connectivity (Fig. 6G). It took increasing the contribution from the top-down input (i.e., to more than 1/3 of the contribution from the stimulus), before the cross-stimulus NC connectivity can be generated by the top-down modulation (Fig. 6H). Thus, this analysis provides further evidence that top-down modulation was not playing a major role in the NC connectivity we observed.

These new results support our original conclusion that network connectivity is the principal mechanism underlying the stability of functional networks.

Public Reviews:

Reviewer #1 (Public Review):

Using multi-region two-photon calcium imaging, the manuscript meticulously explores the structure of noise correlations (NCs) across the mouse visual cortex and uses this information to make inferences about the organization of communication channels between primary visual cortex (V1) and higher visual areas (HVAs). Using visual responses to grating stimuli, the manuscript identifies 6 tuning groups of visual cortex neurons and finds that NCs are highest among neurons belonging to the same tuning group whether or not they are found in the same cortical area. The NCs depend on the similarity of tuning of the neurons (their signal correlations) but are preserved across different stimulus sets - noise correlations recorded using drifting gratings are highly correlated with those measured using naturalistic videos. Based on these findings, the manuscript concludes that populations of neurons with high NCs constitute discrete communication channels that convey visual signals within and across cortical areas.

Experiments and analyses are conducted to a high standard and the robustness of noise correlation measurements is carefully validated. However, the interpretation of noise correlation measurements as a proxy from network connectivity is fraught with challenges. While the data clearly indicates the existence of distributed functional ensembles, the notion of communication channels implies the existence of direct anatomical connections between them, which noise correlations cannot measure.

The traditional view of noise correlations is that they reflect direct connectivity or shared inputs between neurons. While it is valid in a broad sense, noise correlations may reflect shared top-down input as well as local or feedforward connectivity. This is particularly

important since mouse cortical neurons are strongly modulated by spontaneous behavior (e.g. Stringer et al, Science, 2019). Therefore, noise correlation between a pair of neurons may reflect whether they are similarly modulated by behavioral state and overt spontaneous behaviors. Consequently, noise correlation alone cannot determine whether neurons belong to discrete communication channels.

Behavioral modulation can influence the gain of sensory-evoked responses (Niell and Stryker, Neuron, 2010). This can explain why signal correlation is one of the best predictors of noise correlations as reported in the manuscript. A pair of neurons that are similarly gain-modulated by spontaneous behavior (e.g. both active during whisking or locomotion) will have higher noise correlations if they respond to similar stimuli. Top-down modulation by the behavioral state is also consistent with the stability of noise correlations across stimuli. Therefore, it is important to determine to what extent noise correlations can be explained by shared behavioral modulation.

We thank the reviewer for the constructive and positive feedback on our study.

The reviewer acknowledged the quality of our experiments and analysis and stated a concern that the noise correlation can be explained by top-down modulation. We have addressed this concern carefully in the revision, please see the General Response above.

Reviewer #2 (Public Review):

Summary:

This groundbreaking study characterizes the structure of activity correlations over a millimeter scale in the mouse cortex with the goal of identifying visual channels, specialized conduits of visual information that show preferential connectivity. Examining the statistical structure of the visual activity of L2/3 neurons, the study finds pairs of neurons located near each other or across distances of hundreds of micrometers with significantly correlated activity in response to visual stimulation. These highly correlated pairs have closely related visual tuning sharing orientation and/or spatial and/or temporal preference as would be expected from dedicated visual channels with specific connectivity.

Strengths:

The study presents best-in-class mesoscopic-scale 2-photon recordings from neuronal populations in pairs of visual areas (V1-LM, V1-PM, V1-AL, V1-LI). The study employs diverse visual stimuli that capture some of the specialization and heterogeneity of neuronal tuning in mouse visual areas. The rigorous data quantification takes into consideration functional cell groups as well as other variables that influence trial-to-trial correlations (similarity of tuning, neuronal distance, receptive field overlap). The paper convincingly demonstrates the robustness of the clustering analysis and of the activity correlation measurements. The calcium imaging results convincingly show that noise correlations are correlated across visual stimuli and are strongest within cell classes which could reflect distributed visual channels. A simple simulation is provided that suggests that recurrent connectivity is required for the stimulus invariance of the results. The paper is well-written and conceptually clear. The figures are beautiful and clear. The arguments are well laid out and the claims appear in large part supported by the data and analysis results (but see weaknesses).

Weaknesses:

An inherent limitation of the approach is that it cannot reveal which anatomical connectivity patterns are responsible for observed network structure. The modeling results presented, however, suggest interestingly that a simple feedforward architecture

may not account for fundamental characteristics of the data. A limitation of the study is the lack of a behavioral task. The paper shows nicely that the correlation structure generalizes across visual stimuli. However, the correlation structure could differ widely when animals are actively responding to visual stimuli. I do think that, because of the complexity involved, a characterization of correlations during a visual task is beyond the scope of the current study.

An important question that does not seem addressed (but it is addressed indirectly, I could be mistaken) is the extent to which it is possible to obtain reliable measurements of noise correlation from cell pairs that have widely distinct tuning. L2/3 activity in the visual cortex is quite sparse. The cell groups laid out in Figure S2 have very sharp tuning. Cells whose tuning does not overlap may not yield significant trial-to-trial correlations because they do not show significant responses to the same set of stimuli, if at all any time. Could this bias the noise correlation measurements or explain some of the dependence of the observed noise correlations on signal correlations/similarity of tuning? Could the variable overlap in the responses to visual responses explain the dependence of correlations on cell classes and groups?

With electrophysiology, this issue is less of a problem because many if not most neurons will show some activity in response to suboptimal stimuli. For the present study which uses calcium imaging together with deconvolution, some of the activity may not be visible to the experimenters. The correlation measure is shown to be robust to changes in firing rates due to missing spikes. However, the degree of overlap of responses between cell pairs and their consequences for measures of noise correlations are not explored.

Beyond that comment, the remaining issues are relatively minor issues related to manuscript text, figures, and statistical analyses. There are typos left in the manuscript. Some of the methodological details and results of statistical testing also seem to be missing. Some of the visuals and analyses chosen to examine the data (e.g., box plots) may not be the most effective in highlighting differences across groups. If addressed, this would make a very strong paper.

We thank the reviewer for acknowledging the contributions of our study.

We agree with the reviewer that future studies on behaviorally engaged animals are necessary. Although we also agree with the reviewer that behavior studies are out the scope of the current manuscript, we have included additional analysis and discussion on whether and how top-down input would affect the NC connectivity in the revision. Please see the General Response above.

Reviewer #3 (Public Review):

Summary:

Yu et al harness the capabilities of mesoscopic 2P imaging to record simultaneously from populations of neurons in several visual cortical areas and measure their correlated variability. They first divide neurons into 65 classes depending on their tuning to moving gratings. They found the pairs of neurons of the same tuning class show higher noise correlations (NCs) both within and across cortical areas. Based on these observations and a model they conclude that visual information is broadcast across areas through multiple, discrete channels with little mixing across them.

NCs can reflect indirect or direct connectivity, or shared afferents between pairs of neurons, potentially providing insight on network organization. While NCs have been comprehensively studied in neuron pairs of the same area, the structure of these

correlations across areas is much less known. Thus, the manuscripts present novel insights into the correlation structure of visual responses across multiple areas.

Strengths:

The study uses state-of-the art mesoscopic two-photon imaging.

The measurements of shared variability across multiple areas are novel.

The results are mostly well presented and many thorough controls for some metrics are included.

Weaknesses:

I have concerns that the observed large intra-class/group NCs might not reflect connectivity but shared behaviorally driven multiplicative gain modulations of sensory-evoked responses. In this case, the NC structure might not be due to the presence of discrete, multiple channels broadcasting visual information as concluded. I also find that the claim of multiple discrete broadcasting channels needs more support before discarding the alternative hypothesis that a continuum of tuning similarity explains the large NCs observed in groups of neurons.

Specifically:

Major concerns:

(1) Multiplicative gain modulation underlying correlated noise between similarly tuned neurons

(1a) The conclusion that visual information is broadcasted in discrete channels across visual areas relies on interpreting NC as reflecting, direct or indirect connectivity between pairs, or common inputs. However, a large fraction of the activity in the mouse visual system is known to reflect spontaneous and instructed movements, including locomotion and face movements, among others. Running activity and face movements are some of the largest contributors to visual cortex activity and exert a multiplicative gain on sensory-evoked responses (Niell et al, Stringer et al, among others). Thus, trial-by-fluctuations of behavioral state would result in gain modulations that, due to their multiplicative nature, would result in more shared variability in cotuned neurons, as multiplication affects neurons that are responding to the stimulus over those that are not responding (see Lin et al, Neuron 2015 for a similar point).

As behavioral modulations are not considered, this confound affects most of the conclusions of the manuscript, as it would result in larger NCs the more similar the tuning of the neurons is, independently of any connectivity feature. It seems that this alternative hypothesis can explain most of the results without the need for discrete broadcasting channels or any particular network architecture and should be addressed to support its main claims.

(1b) In Figure 5 the observations are interpreted as evidence for NCs reflecting features of the network architecture, as NCs measured using gratings predicted NC to naturalistic videos. However, it seems from Figure 5 A that signal correlations (SCs) from gratings had non-zero correlations with SCs during naturalistic videos (is this the case?). Thus, neurons that are cotuned to gratings might also tend to be coactivated during the presentation of videos. In this case, they are also expected to be susceptible to shared behaviorally driven fluctuations, independently of any circuit architecture as explained before. This alternative interpretation should be addressed before concluding that these measurements reflect connectivity features.

We thank the reviewer for acknowledging the contributions of our study.

The reviewer suggested that gain modulation might be interfering with the interpretation of the NC connectivity. We have addressed this issue in the General Response above.

Here, we will elaborate on one additional analysis we performed, in case it might be of interest. We carried out multiplicative gain modeling by implementing an established method (Goris et al. 2014 Nat Neurosci) on our dataset. We were able to perform the modeling work successfully. However, we found that it is not a suitable model for explaining the current dataset because the multiplicative gain induced a negative correlation. This seemed odd but can be explained. First, top-down input is not purely multiplicative but rather both additive and multiplicative. Second, the top-down modulation is high dimensional. Third, the firing rate of layer 2/3 mouse visual cortex neurons is lower than the firing rates for non-human primate recordings used in the development of the method (Goris et al. 2014 Nat Neurosci). Thus, we did not pursue the model further. We just mention it here in case the outcome might be of interest to fellow researchers.

(2) Discrete vs continuous communication channels

(2a) One of the author's main claims is that the mouse cortical network consists of discrete communication channels. This discreteness is based on an unbiased clustering approach to the tuning of neurons, followed by a manual grouping into six categories in relation to the stimulus space. I believe there are several problems with this claim. First, this clustering approach is inherently trying to group neurons and discretise neural populations. To make the claim that there are 'discrete communication channels' the null hypothesis should be a continuous model. An explicit test in favor of a discrete model is lacking, i.e. are the results better explained using discrete groups vs. when considering only tuning similarity? Second, the fact that 65 classes are recovered (out of 72 conditions) and that manual clustering is necessary to arrive at the six categories is far from convincing that we need to think about categorically different subsets of neurons. That we should think of discrete communication channels is especially surprising in this context as the relevant stimulus parameter axes seem inherently continuous: spatial and temporal frequency. It is hard to motivate the biological need for a discretely organized cortical network to process these continuous input spaces.

(2b) Consequently, I feel the support for discrete vs continuous selective communication is rather inconclusive. It seems that following the author's claims, it would be important to establish if neurons belong to the same groups, rather than tuning similarity is a defining feature for showing large NCs.

Thanks for pointing this out so that we can clarify.

We did not mean to argue that the tuning of neurons is discrete. Our conclusions are not dependent on asserting a particular degree of discreteness. We performed GMM clustering to label neurons with an identity so that we could analyze the NC connectivity structure with a degree of granularity supported by the data. Our analysis suggested that communication happens within a class, rather than through mixed classes. We realized that using the term “discrete” may be confusing. In the revised text we used the term “unmixed” or “non-mixing” instead to emphasize that the communication happens between neurons belonging to the same tuning cluster, or class.

However, we do see how the question of discreteness among classes might be interesting to readers. To provide further information, we have included a new Fig. S2 to visualize the GMM classes using t-SNE embedding.

Finally, as stated in point 1, the larger NCs observed within groups than across groups might be due to the multiplicative gain of state modulations, due to the larger tuning similarity of the neurons within a class or group.

We have addressed this issue in the General Response above and the response to comment (1).

Recommendations for the authors:

Reviewing Editor (Recommendations For The Authors):

A general recommendation discussed with the reviewers is to make use of behavioural recording to assess whether shared behaviourally driven modulations can explain the observed relation between SC and NC, independently of the network architecture. Alternatively, a simulation or model might also address this point as well as the possibility that the relation of SC and NC might be also independent of network architecture given the sparseness of the sensory responses in L2/3.

We have addressed this in the General Response above.

Broadly speaking, inferring network architecture based on NCs is extremely challenging. Consequently, the study could also be substantially improved by reframing the results in terms of distributed co-active ensembles without insinuation of direct anatomical connectivity between them.

We agree that the inferring network architecture based on NCs is challenging. The current study has revealed some principles of functional networks measured by NCs, and we showed that cross-stimulus NC connectivity provides effective constraints to network modeling. We are explicit about the nature of NCs in the manuscript. For example, in the Abstract, we write “to measure correlated variability (i.e., noise correlations, NCs)”, and in the Introduction, we write “NCs are due to connectivity (direct or indirect connectivity between the neurons, and/or shared input)”. We are following conventions in the field (e.g., Sporns 2016; Cohen and Kohn 2011).

Notice also that the abstract or title should make clear that the study was made in mice.

Sorry for the confusion, we now clearly state the study was carried out in mice in the Abstract and Introduction.

Reviewer #1 (Recommendations For The Authors):

The manuscript presents a meticulous characterization of noise correlations in the visual cortical network. However, as I outline in the public review, I think the use of noise correlations to infer communication channels is problematic and I urge the authors to carefully consider this terminology. Language such as "strength of connections" (Figure 4D) should be avoided.

We now state in the figure legend that the plot in Fig. 4D shows the average NC value.

My general suggestion to the authors, which primarily concerns the interpretation of analyses in Figures 4-6, is to consider the possible impact of shared top-down modulation on noise correlations. If behavioral data was recorded simultaneously (e.g. using cameras to record face and body movements), behavioral modulation should be considered alongside signal correlation as a possible factor influencing NCs.

We have addressed this issue in the General Response above.

I may be misunderstanding the analysis in Figure 4C but it appears circular. If the fraction of neurons belonging to a particular tuning group is larger, then the number of in-group high NC pairs will be higher for that group even if high NC pairs are distributed randomly. Can you please clarify? I frankly do not understand the analysis in Figure 4D and it is unclear to me how the analyses in Figure 4C-D address the hypotheses depicted in the cartoons.

Sorry for the confusion, we have clarified this in the Fig. 4 legend.

Each HVA has a SFTF bias (Fig. 1E,F; Marshel et al., 2011; Andermann et al., 2011; Vries et al., 2020). Each red marker on the graph in Fig. 4C is a single V1-HVA pair (blue markers are within an area) for a particular SFTF group (Fig. 1). The x-axis indicates the number of high NC pairs in the SFTF group in the V1-HVA pair divided by the total number of high NC pairs per that V1-HVA pair (summed over all SFTF groups). The trend is that for HVAs with a bias towards a particular SFTF group, there are also more high NC pairs in that SFTF group, and thus it is consistent with the model on the right side. This is not circular because it is possible to have a SFTF bias in an HVA and have uniformly low NCs. The reviewer is correct that a random distribution of high NCs could give a similar effect, which is still consistent with the model: that the number of high NC pairs (and not their specific magnitudes) can account for SFTF biases in HVAs.

To contrast with that model, we tested whether the average NC value for each tuning group varies. That is, can a small number of very high NCs account for SFTF biases in HVAs? That is what is examined in Fig. 4D. We found that the average NC value does not account for the SFTF biases. Thus, the SFTF biases were not related to the modulation in NC (i.e., functional connection strength).

I found the discussion section quite odd and did not understand the relevance of the discussion of the coefficient of variation of various quantities to the present manuscript. It would be more useful to discuss the limitations and possible interpretations of noise correlation measurements in more detail.

We have revised the discussion section to focus on interpreting the results of the current study and comparing them with those of previous studies.

Figure 3B: please indicate what the different colors mean - I assume it is the same as Figure 3A but it is unclear.

We added text to the legend for clarification.

Typos: Page 7: "direct/indirection wiring", Page 11: "pooled over all texted areas"

We have fixed the typos.

Reviewer #2 (Recommendations For The Authors):

The significance of the results feels like it could be articulated better. The main conclusion is that V1 to HVA connections avoid mixing channels and send distinctly tuned information along distinct channels - a more explicit description of what this functional network understanding adds would be useful to the reader.

Thanks for the suggestion. We have edited the introduction section and the discussion section to make the take-home message more clear.

Previous studies with anatomical data already indicate distinctly tuned channels - several of which the authors cite - although inconsistently:

- Kim et al 2018 <https://doi.org/10.1016/j.neuron.2018.10.023>
- Glickfeld et al., 2013 (cited)
- Han et al., 2022 (cited)
- Han and Bonin 2023 (cited)

Thanks for the suggestion, we now cite the Kim et al. 2018 paper.

I think the information you provide is valuable - but the value should be more clearly spelled out - This section from the end of the discussion for example feels like abdicates that responsibility:

"In summary, mesoscale two-photon imaging techniques open up the window of cellular-resolution functional connectivity at the system level. How to make use of the knowledge of functional connectivity remains unclear, given that functional connectivity provides important constraints on population neuron behavior."

A discussion of how the results relate to previous studies and a section on the limitations of the study seems warranted.

Thanks for the suggestion, we have extensively edited the discussion section to make the take-home message clear and discuss prior studies and limitations of the present study.

Details:

Analyses or simulations showing that the dependency of correlations on similarity of tuning is not an artifact of how the data was acquired is in my mind missing and if that is the case it is crucial that this be addressed.

At each step of data analysis, we performed control analysis to assess the fidelity of the conclusion. For example, on the spike train inference (Fig. S4), GMM clustering (Fig. S1), and noise correlation analysis (Figs. 2, S5).

None of the statistical testing seems to use animals as experimental units (instead of neurons). This could over-inflate the significance of the results. Wherever applicable and possible, I would recommend using hierarchical bootstrap for testing or showing that the differences observed are reproducible across animals.

We analyzed the tuning selectivity of HVAs (Fig. 1F) using experimental units, rather than neurons. It is very difficult to observe all tuning classes in each experiment, so pooling neurons across animals is necessary for much of the analysis. We do take care to avoid overstating statistical results, and we show the data points in most figure to give the reader an impression of the distributions.

Page 2. "The number of neurons belonged to the six tuning groups combined: V1, 5373; LM, 1316; AL, 656; PM, 491; LI, 334." Yet the total recorded number of neurons is 17,990. How neurons were excluded is mentioned in Methods but it should be stated more explicitly in Results.

We have added text in the Fig. 1 legend to direct the audience to the Methods section for information on the exclusion / inclusion criteria.

Figure 1C, left. I don't understand how correlation is the best way to quantify the consistency of class center with a subset of data. Why not use for example as the mean square error. The logic underlying this analysis is not explained in Methods.

Sorry for the confusion, we have clarified this in the Methods section.

We measured the consistency of the centers of the Gaussian clusters, which are 45-dimensional vectors in the PC dimensions. We measured the Pearson correlation of Gaussian center vectors independently defined by GMM clustering on random subsets of neurons. We found the center of the Gaussian profile of each class was consistent (Fig. 1C). The same class of different GMMs was identified by matching the center of the class.

Figure 1E. There are statements in the text about cell groups being more represented in certain visual areas. These differences are not well represented in the box plots. Can't the individual data points be plotted? I have also not found the description and results of statistical testing for these data.

We have replotted the figure (now Fig. 1F) with dot scatters which show all of the individual experiments.

Figure 2A, right, since these are paired data, I am not quite sure why only marginal distributions are shown. It would be interesting to know the distributions of correlations that are significant.

This is only for illustration showing that NCs are measurable and significantly different from zero or shuffled controls. The distribution of NCs is broad and has both positive and negative values. We are not using this for downstream analysis.

Figure 4A, I wonder if it would not be better to concentrate on significant correlations.

We focused on large correlation values rather than significant values because we wanted to examine the structure of “strongly connected” neuron pairs. Negative and small correlation values can be significant as well. Focusing on large values would allow us to generate a clear interpretation.

Figure 4B, 'Mean strength of connections' which I presume mean correlations is not defined anywhere that I can see.

I believe the reviewer means Fig. 4D. It means the average NC value. We have edited the figure legend to add clarity.

Figure 4F, a few words explaining how to understand the correlation matrix in text or captions would be helpful.

Sorry for the confusion, we have clarified this part in figure legend for Fig. 4F.

Page 5, right column: Incomplete sentence: "To determine whether it is the number of high NC pairs or the magnitude of the NCs,".

We have edited this sentence.

Page 5, right column: "Prior findings from studies of axonal projections from V1 to HVAs indicated that the number of SF-TF-specific boutons -rather than the strength of boutons- contribute to the SF-TF biases among HVAs (Glickfeld et al., 2013)." Glickfeld et al. also reported that boutons with tuning matched to the target area showed stronger peak dF/F responses.

Thank you. We have revised this part accordingly.

Page 9, the Discussion and Figure 7 which situates the study results in a broader context is welcome and interesting, but I have the feeling that more words should be spent explaining the figure and conceptual framework to a non-expert audience. I am a bit at a loss about how to read the information in the figure.

Sorry for the confusion, we have added an explanation about this section (page 10, right column).

As far as I can see, data availability is not addressed in the manuscript. The data, code to analyze the data and generate the figures, and simulation code should be made available in a permanent public repository. This includes data for visual area mapping, calcium imaging data, and any data accessory to the experiments.

We have stated in the manuscript that code and data are available upon request. We regularly share data with no conditions (e.g., no entitlement to authorship), and we often do so even prior to publication.

The sex of the mice should be indicated in Figure T1.

The sex of the mice was mixed. This is stated in the Methods section.

Methods:

Section on statistical testing, computation of explained variance missing, etc. I feel many analyses are not thoroughly described.

Sorry for the confusion, we have improved our method section.

Signal correlation (similarity between two neurons' average responses to stimuli) and its relation to noise correlation is not formally defined.

We have included the definition of signal correlation in the Methods.

Number of visual stimulation trials is not stated in Methods. Only stated figure caption.

The number of visual stimulus trials is provided in the last paragraph of the Methods section (Visual Stimuli).

Fix typos: incorrect spelling, punctuation, and missing symbols (e.g. closing parentheses).

We have carefully examined the spelling, punctuation, and grammar. We have corrected errors and we hope that none remain.

Why use intrinsic imaging to locate retinotopic boundaries in mice already expressing GCaMP6s?

We agree with the reviewer that calcium imaging of visual cortex can be used to identify the visual cortex.

It is true that areas can be mapped using the GCaMP signals. That is not our preferred approach. Using intrinsic imaging to define the boundary between V1 and HVAs has been a well refined routine in our lab for over a decade. It is part of our standard protocol. One advantage is that the data (from intrinsic signals) is of the same nature every time. This enables us to use the same mapping procedure no matter what reporters mice might be expressing (and the pattern, e.g., patchy or restricted to certain cell types).

Reviewer #3 (Recommendations For The Authors):

The possibility that larger intra-group NCs observed simply reflect a multiplicative gain on cotuned neurons could be addressed using pupil and/or face recordings: Does pupil size or facial motion predict NCs and if factored out, does signal correlation still predict NCs?

Perhaps a variant of the network model presented in Figure 6 with multiplicative gain could also be tested to investigate these issues.

We have addressed this issue in general response.

Here, we will elaborate on one additional analysis we performed, in case it might be of interest. We carried out multiplicative gain modeling by implementing an established method (Goris et al. 2014 Nat Neurosci) on our dataset. We were able to perform the modeling work successfully. However, we found that it is not a suitable model for explaining the current dataset because the multiplicative gain induced a negative correlation. This seemed odd but can be explained. First, top-down input is not purely multiplicative but rather both additive and multiplicative. Second, the top-down modulation is high dimensional. Third, the firing rate of layer 2/3 mouse visual cortex neurons is lower than the firing rates for non-human primate recordings used in the development of the method (Goris et al. 2014 Nat Neurosci). Thus, we did not pursue the model further. We just mention it here in case the outcome might be of interest to fellow researchers.

Similarly further analyses can be done to strengthen support for the claims that the observed NCs reflect discrete communication channels. A direct test of continuous vs categorical channels would strengthen the conclusions. One possible analysis would be to compare pairs with similar tuning (same SC) belonging to the same or different groups.

Thanks for pointing this out so that we can clarify.

We did not mean to argue that the tuning of neurons is discrete. Our conclusions are not dependent on asserting a particular degree of discreteness. We performed GMM clustering to label neurons with an identity so that we could analyze the NC connectivity structure with a degree of granularity supported by the data. Our analysis suggested that communication happens within a class, rather than through mixed classes. We realized that using the term “discrete” may be confusing. In the revised text we used the term “unmixed” or “non-mixing” instead to emphasize that the communication happens between neurons belonging to the same tuning cluster, or class.

However, we do see how the question of discreteness among classes might be interesting to readers. To provide further information, we have included a new Fig. S2 to visualize the GMM classes using t-SNE embedding.

I also found many places where the manuscript needs clarification and /or more methodological details:

- *How many times was each of the stimulus conditions repeated? And how many times for the two naturalistic videos? What was the total duration of the experiments?*

The number of visual stimulus trials is provided in the last paragraph of the Methods section entitled *Visual Stimuli*. About 15 trials were recorded for each drifting grating stimulus, and about 20 trials were recorded for each naturalistic video.

- *Typo: Suit2p should be Suite2p (section Calcium image processing - Methods).*

We have fixed the typo.

- *What do the error bars in Figure 1E represent? Differences in group representation across areas from Figure 1E are mentioned in the text without any statistical testing.*

We have revised the Figure 1E (current Fig. 1F), and we now show all data points.

- *The manuscript would benefit from a comparison of the observed area-specific tuning biases across areas (Figure 1E and others) with the previous literature.*

We have included additional discussion on this in the last paragraph of the section entitled *Visual cortical neurons form six tuning groups*.

- *Why are inferred spike trains used to calculate NCs? Why can't dF/F be used? Do the results differ when using dF/F to calculate NC? Please clarify in the text.*

We believe inferred spike trains provide better resolution and make it easier to compare with quantitative values from electrical recordings. Notice that NC values computed using dF/F can be much larger than those computed by inferred spike trains. For example, see Smith & Hausser 2010 Nat Neurosci. Supplementary Figure S8.

- *The sentence seems incomplete or unclear: "That is, there are more high NC pairs that are in-group." Explicit vs what?*

We have revised this sentence.

- *Figure 1E is unclear to me. What is being plotted? Please add a color bar with the metric and the units for the matrix (left) and in the tuning curves (right panels). If the Y and X axes represent the different classes from the GMM, why are there more than 65 rows? Why is the matrix not full?*

We have revised this figure. Fig. 1D is the full 65 x 65 matrix. Fig. 1F has small 3x3 matrices mapping the responses to different TF and SF of gratings. We hope the new version is clearer.

- *How are receptive fields defined? How are their long and short axes calculated? How are their limits defined when calculating RF overlap?*

We have added further details in the Methods section entitled “*Receptive field analysis*”.

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